

A locking-free finite element formulation and reduced models for geometrically exact Kirchhoff rods

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Abstract

In this work, we suggest a locking-free geometrically exact finite element formulation incorporating the modes of axial tension, torsion and bending of thin Kirchhoff beams with arbitrary initial curvatures. The proposed formulation has been designed in order to represent general load cases and three-dimensional problem settings in the geometrically nonlinear regime of large deformations. From this comprehensive theory, we not only derive a general beam model but also several reduced formulations, which deliver accurate solutions for special problem classes concerning the beam geometry and the external loads. The advantages of these reduced models arise for example in terms of simplified finite element formulations, less degrees of freedom per element and consequently a higher computational efficiency of the corresponding numerical models. A second core topic of this publication is the treatment of membrane locking, which is a locking phenomenon predominantly occurring in highly slender curved structures, thus, exactly in the prime field of application for Kirchhoff theories. In order to address the membrane locking effect, we will propose a new interpolation strategy for the axial strain field and compare this method with common approaches such as Assumed Natural Strains (ANS) or reduced integration. The effectiveness of this method as well as the consistency and accuracy of the general finite element formulation and the reduced beam models will be illustrated with selected numerical examples.

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1. Introduction

For many years, the design of geometrically exact beam elements according to the Cosserat theory (see e.g. [1]) has essentially been restricted to shear-deformable beam formulations. The starting point for these developments can be traced back to the important works of Reissner [2,3] and Simo [4,5]. Crisfield and Jelenic were the first ones who recognized the special significance of objectivity and path-independence in the context of geometrically exact beams, see [6,7]. They have shown the lack of these essential properties for all existing geometrically exact formulations at that time and developed a strain-invariant and path-independent interpolation strategy for the rotational degrees of freedom of nonlinear beam elements. Many other authors followed this example and developed conceptually

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different interpolation methods, still preserving objectivity and path-independence (see e.g. [8–15]). In his recent contributions [11,16], Romero pointed out the excellent performance of geometrically exact beam formulations of Simo–Reissner type, especially when designed according to the principles presented in [6]. Furthermore, [17] presents a recent overview concerning the development of different types of geometrically exact Simo–Reissner beams and possible difficulties.

However, state-of-the-art shear-free beam elements according to the Kirchhoff theory of thin rods [18,19] have not reached these outstanding properties of geometrically exact Simo–Reissner beams so far. In the remainder of this work, the notion “shear-free” is equivalent to “vanishing shear strains”, but, of course, not to “vanishing shear stresses” for general load states. In the early works [20,21], numerical investigations and comparisons of the standard Kirchhoff beam elements existing at that time have been made. Accordingly, many of these first representatives showed a poor numerical accuracy, a fact that amongst others was attributed to the non-objective nature of these formulations. Different strategies in order to preserve objectivity and improve the accuracy of such elements have been developed in the following years. However, most of these strategies were limited to plane problems and simple geometries like circular arcs or based on straight “framework” or “facet” element approximations. It has been shown in [22] that these “framework” or “facet” element approximations, when applied to Kirchhoff beam formulations, may lead to a deterioration of the numerical performance as compared with corresponding geometrically exact approaches.

A further critical issue relevant in the context of thin Kirchhoff beams is membrane locking, a locking phenomenon given distinction to by Stolarski and Belytschko [23]. In general, membrane locking denotes the inability of curved structural elements, e.g. beams or shells, to represent the inextensibility constraint of vanishing membrane/axial strains. For thin Kirchhoff beams, [24] was one of the first contributions in which this effect was investigated by relating the beam slenderness ratio to the condition number of the stiffness matrix, but without explicitly using the term locking. Diverse methods have been proposed in the literature in order to avoid membrane locking of Kirchhoff rods. Amongst others, these are the approaches of reduced/selective integration (see e.g. [25–27]), assumed strains based on the Hu–Washizu functional (see e.g. [28–30]), assumed stresses based on the Hellinger–Reissner functional (see e.g. [31,25]) or penalty relaxation/stabilization techniques in combination with membrane correction factors (see e.g. [32,33]). A historic overview concerning the development of Kirchhoff beams in general and the key issues of objectivity and membrane locking in particular is given in the recent works [22,34].

Almost all works on Kirchhoff beams presented above are restricted to two-dimensional problems. In general, there exist only very few three-dimensional finite element formulations for thin Kirchhoff rods, as e.g. [35–37]. Furthermore, these formulations only treat the special case of beams with circular cross-sections and a straight, stress-free reference configuration, which will be denoted as “isotropic bending” case later on in this work. For these reasons, Armero recently developed Kirchhoff beam elements for arbitrarily curved initial geometries that guarantee the fundamental properties of objectivity and geometrical exactness [22,34]. The corresponding beam elements only cover the case of infinitesimal deformations and have to be considered as a first step towards general large-deformation Kirchhoff beam elements. In [38], also some first steps towards a geometrically nonlinear Kirchhoff beam element have been made, however, without addressing the important issues of objectivity and membrane locking.

This lack of shear-free beam elements with properties comparably excellent as those of geometrically exact Simo–Reissner beam elements and our need for such high quality elements capable of covering the challenging field of very high slenderness ratios was our motivation for the development of a geometrically exact, objective, three-dimensional, large-deformation element formulation for thin Kirchhoff rods that accounts for arbitrarily curved initial geometries. Within this contribution, the notion “large-deformation” primarily refers to large translations and large rotations. Although geometrically exact beam formulations are by definition internally consistent regardless of the magnitude of displacements, rotations and strains, it has already been argued in [6,39] that an application of geometrically exact beams to problems beyond the small-strain regime is at least questionable. In our earlier work [39], we have already presented a detailed derivation of this element formulation and have shown analogies to the geometrically exact Simo–Reissner theory. Furthermore, we have already investigated the crucial topics of objectivity and path-independence, the properties of the applied Hermite interpolation and the question of proper reference triads required for this formulation.

Within the current work, we want to further improve the resulting element formulation in particular with regard to membrane locking. In [39], we considered beams with moderate slenderness ratios, i.e. geometries in which membrane locking is known not to be too severe. Now, we want to focus on beams of extreme slenderness, a proceeding that is motivated twofold: On the one hand, there exist several important fields of application, such as biopolymer networks

(see e.g. [40,41]) or fibers in high-tensile webbings (see e.g. [42]), consisting of highly slender structures that can be best modeled as 1D-continua. On the other hand, it is precisely the regime of very high slenderness ratios for which the shear-free theory of thin Kirchhoff rods is tailored. Since for such geometries membrane locking indeed plays a crucial role, we will propose a new method, denoted as Minimal Constraint Strains (MCS), in order to avoid this phenomenon.

This method, based on the principle of assumed strains, is of a very general nature, a fact that lays the foundation for a possible transfer to other locking phenomena occurring in geometrically nonlinear structures. It will be shown that this method effectively eliminates the shortfalls resulting from membrane locking without a noteworthy increase of the numerical effort. Furthermore, we will compare the new MCS method with existing approaches such as Assumed Natural Strains (ANS) or reduced/selective integration. It will turn out that the effectiveness of the classical ANS method is in general not fully transferable from geometrically linear to nonlinear problems, while the functionality of the MCS approach is independent of the degree of geometrical nonlinearity.

A second core topic of this work is the derivation of reduced beam models, i.e. beam models that are only valid for special problem classes concerning the beam geometry and the external loads. In addition to the full model, we will derive these reduced formulations in a consistent manner starting with the general theory. The advantages of such reduced models are simplified element formulations, less degrees of freedom per element and a higher computational efficiency, to name only a few. The extent of validity of these reduced models will be stringently defined and their properties will be compared with the properties of existing reduced models, such as cables (see e.g. [43,44]) or torsion-free Simo–Reissner beams (see e.g. [45]). Of course, the development of these reduced formulations is not only warrantable from a theoretical point of view. Firstly, the geometries and external loads in fields of applications such as the aforementioned actually fulfill the requirements in order to apply these reduced models. Secondly, the large number of degrees of freedom and the long physically relevant time scales typical for systems such as biopolymer networks underline the importance of very compact and efficient simulation tools.

The content of this work is structured as follows: In Section 2, the main steps in the derivation of the general Kirchhoff beam formulation including objective deformation measures, kinematic and constitutive relations and the weak form of the equilibrium equations will be presented. One of the crucial factors in the element design is the definition of a so-called reference triad field, whose proper interpolation is essential for the invariance of the strain measures. While the corresponding reference triad applied to the finite element formulation will be introduced in Section 2.3, an alternative reference triad definition, which enables a convenient analytical treatment of Kirchhoff beam problems, will be presented in Section 2.4. Starting with the general beam formulation, we will consistently derive reduced beam models in Section 3. After introducing the main aspects of spatial discretization in Section 4, we investigate the issue of membrane locking in the scope of slender Kirchhoff beams in Section 5. There, we will present our new MCS method for the avoidance of membrane locking and compare it with existing approaches such as ANS or reduced/selective integration. Finally, in Section 6, the effectivity of this method as well as the consistency and accuracy of the general and the reduced models will be verified by means of suitable numerical examples.

2. Theory of geometrically exact Kirchhoff beams

In this section, the basics of the developed geometrically exact Kirchhoff beam formulation will be presented. The corresponding theory will serve as a starting point for the derivations in Sections 3 and 5. More details of the derivation including a comparison with the Simo–Reissner beam theory can be found in [39].

2.1. Kinematics

In order to describe material and spatial configurations, we introduce material and spatial right-handed inertial Cartesian frames $\mathbf{E}_i$ and $\mathbf{e}_i$, respectively, which for simplicity should coincide, i.e. $\mathbf{E}_i = \mathbf{e}_i$ for $i = 1, 2, 3$. In the initial (unstressed) configuration of the beam, the line of cross-section centroids, in the following denoted as beam centerline, is described by the space curve $s \rightarrow \mathbf{r}_0(s) \in \mathbb{R}^3$. Throughout this work, the index 0 refers to the initial configuration. Furthermore, $s \in [0, L] \subset \mathbb{R}$ is an arc-length parametrization of the curve and $L \in \mathbb{R}$ the beam length in the initial configuration. The description of the initial configuration is completed by a field of right-handed orthonormal triads, denoted as material triads and consisting of the base vectors $s \rightarrow \mathbf{g}_{01}(s), \mathbf{g}_{02}(s), \mathbf{g}_{03}(s) \in \mathbb{R}^3$,

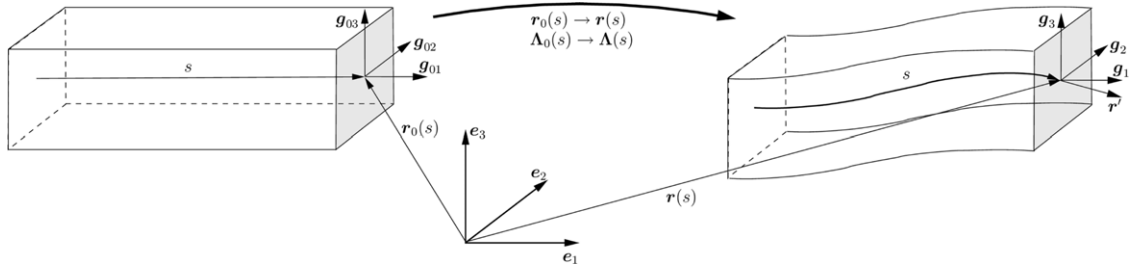


Fig. 1. Kinematic quantities defining the initial and deformed configurations of the beam.

which are attached to the beam cross-sections. The first base vector $\mathbf{g}_{01}$ is identical to the initial unit tangential vector, i.e.

$$\mathbf{g}_{01}(s) \equiv \mathbf{r}'_0(s) \quad \text{with} \quad \|\mathbf{r}'_0(s)\| = 1, \quad (1)$$

and the base vectors $\mathbf{g}_{02}(s)$ and $\mathbf{g}_{03}(s)$ coincide with the principal axes of inertia of the cross-section. Throughout this work, the prime $(\cdot)' = \frac{d}{ds}(\cdot)$ denotes the derivative with respect to s , and $\|(\cdot)\|$ is the Euclidean norm in $\mathbb{R}^3$. Finally, the orthogonal transformation $s \rightarrow \mathbf{A}_0(s) \in SO(3)$, with $SO(3)$ representing the Special Orthogonal group of 3D rotations with the properties $\det(\mathbf{A}) = 1$ and $\mathbf{A}\mathbf{A}^T = \mathbf{I}$, rotates the global frame $\mathbf{E}_i$ onto the initial local frame $\mathbf{g}_{0i}(s)$ as follows:

$$\mathbf{g}_{0i}(s) = \mathbf{A}_0(s)\mathbf{E}_i(s) \quad \text{for } i = 1, 2, 3 \quad \text{with} \quad \mathbf{A}_0(s) = \mathbf{g}_{0j}(s) \otimes \mathbf{E}_j. \quad (2)$$

Throughout this contribution, a repeated Latin index implies the summation over three components 1, 2, 3. In a similar fashion, the deformed configuration is defined by the quantities $\mathbf{r}(s)$ and $\mathbf{A}(s)$. The kinematic quantities representing the initial and deformed configurations of the beam are illustrated in Fig. 1. The theory presented in the following shall be restricted to the case of vanishing shear strains. The latter can be assumed as a sensible approximation in case of highly slender beams (see e.g. [19]). Thereto, we introduce the constraint that the beam cross-sections must remain perpendicular to the unit tangent vector $\check{\mathbf{t}}(s) := \mathbf{t}(s)/\|\mathbf{t}(s)\|$, with $\mathbf{t}(s) := \mathbf{r}'(s)$, to the beam centerline:

$$\mathbf{g}_2(s) \cdot \check{\mathbf{t}}(s) \equiv 0 \quad \text{and} \quad \mathbf{g}_3(s) \cdot \check{\mathbf{t}}(s) \equiv 0 \quad \text{or} \quad \mathbf{g}_1(s) \equiv \check{\mathbf{t}}(s). \quad (3)$$

Next, we want to choose a parametrization consisting of 4 degrees of freedom $(\mathbf{r}(s), \varphi(s))$, with $s \rightarrow \varphi(s) \in \mathbb{R}$, which fulfills this Kirchhoff constraint in a strong manner. Thereto, we introduce a field of local right-handed orthonormal triads $s \rightarrow \mathbf{g}_{ref,1}(s), \mathbf{g}_{ref,2}(s), \mathbf{g}_{ref,3}(s) \in \mathbb{R}^3$, denoted as reference triads and characterized by the conditions:

$$\begin{aligned} \mathbf{g}_{ref,1}(s) &= \check{\mathbf{t}}(s) = \frac{\mathbf{r}'(s)}{\|\mathbf{r}'(s)\|}, & \mathbf{g}_{ref,2}(s) &= \mathbf{f}_2(\mathbf{r}(s), \mathbf{r}'(s), \mathbf{r}''(s)) \quad \text{and} \\ \mathbf{g}_{ref,3}(s) &= \mathbf{f}_3(\mathbf{r}(s), \mathbf{r}'(s), \mathbf{r}''(s)). \end{aligned} \quad (4)$$

The first base vector $\mathbf{g}_{ref,1}(s)$ of the reference triad equals the unit tangent vector to the beam centerline and the two base vectors $\mathbf{g}_{ref,2}(s)$ and $\mathbf{g}_{ref,3}(s)$ may be arbitrary but well-defined functions $\mathbf{f}_2$ and $\mathbf{f}_3$, which solely depend on the primary variable $\mathbf{r}(s)$ and its derivatives. Thus, the reference triad field is part of the solution and cannot be determined in advance. The material triad can now be generated via a rotation of the reference triad with respect to the tangent:

$$\begin{aligned} \mathbf{g}_1(s) &= \mathbf{g}_{ref,1}(s), & \mathbf{g}_2(s) &= \mathbf{g}_{ref,2}(s) \cos \varphi(s) + \mathbf{g}_{ref,3}(s) \sin \varphi(s), \\ \mathbf{g}_3(s) &= \mathbf{g}_{ref,3}(s) \cos \varphi(s) - \mathbf{g}_{ref,2}(s) \sin \varphi(s). \end{aligned} \quad (5)$$

Thus, the fourth degree of freedom $\varphi(s)$ can be interpreted as the relative rotation angle between the reference and the material triad at a certain position s . The reference triad definition applied within this work and the corresponding analytical expression for the functions $\mathbf{f}_2$ and $\mathbf{f}_3$ will be presented in Section 2.3. However, all derivations in the remainder of this section and in Section 2.2 are valid for arbitrary reference triad definitions. Since an analytical expression for the variation of the constrained base vectors $\mathbf{g}_i(s)$ will be needed in the next section, we want to

introduce the constrained rotation vector variation $\delta\boldsymbol{\theta}(s)$. The latter is defined implicitly by the relation

$$\delta\mathbf{g}_i(s) = \delta\boldsymbol{\theta}(s) \times \mathbf{g}_i(s). \quad (6)$$

With the help of (3), the following expression for $\delta\boldsymbol{\theta}(s)$ that obeys the Kirchhoff constraint can be derived:

$$\delta\boldsymbol{\theta}(s) = \delta\alpha(s)\check{\mathbf{t}}(s) + \frac{\mathbf{r}'(s) \times \delta\mathbf{r}'(s)}{\|\mathbf{r}'(s)\|^2}. \quad (7)$$

Here the quantity $s \rightarrow \delta\mathbf{r}(s) \in \mathfrak{R}^3$ describes kinematically admissible variations of the variable $\mathbf{r}(s)$. We will not specify the quantity $s \rightarrow \delta\alpha(s) \in \mathfrak{R}$ any further, since the component of the spin vector $\delta\boldsymbol{\theta}(s)$ that is parallel to $\check{\mathbf{t}}$ is not determined by the Kirchhoff constraint. Instead, $\delta\alpha(s)$ will be regarded as independent variational degree of freedom.

2.2. Deformation measures, constitutive relations and weak form

In order to shorten notation, the argument (s) will often be omitted in the following derivations. Throughout this work, we assume that the considered beam shall be loaded with distributed external forces $\tilde{\mathbf{f}}$ and moments $\tilde{\mathbf{m}}$ per unit length. In this case, the strong form of the static equilibrium equations (see e.g. [1,3] or [4]) is represented by

$$\begin{aligned} \mathbf{f}' + \tilde{\mathbf{f}} &= \mathbf{0}, \\ \mathbf{m}' + \mathbf{r}' \times \mathbf{f} + \tilde{\mathbf{m}} &= \mathbf{0}. \end{aligned} \quad (8)$$

The common definitions of the spatial force stress resultants $\mathbf{f}$ and the spatial moment stress resultants $\mathbf{m}$ can for example be found in [4]. In order to eliminate the shear forces $\mathbf{f}_\perp$, which provide no work contribution in the shear-free case, we split the force vector into a component $\mathbf{f}_\parallel$ parallel and a component $\mathbf{f}_\perp$ perpendicular to the tangent:

$$\mathbf{f} = \mathbf{f}_\parallel + \mathbf{f}_\perp = f_\parallel \check{\mathbf{t}} + \mathbf{f}_\perp. \quad (9)$$

After eliminating $\mathbf{f}_\perp$ from (8), we get the following system of four differential equations in the unknowns $\mathbf{r}$ and φ :

$$\begin{aligned} \check{\mathbf{t}}^T (\mathbf{m}' + \tilde{\mathbf{m}}) &= 0, \\ \mathbf{f}'_\parallel + \left[\frac{\mathbf{r}'}{\|\mathbf{r}'\|^2} \times (\mathbf{m}' + \tilde{\mathbf{m}}) \right]' + \tilde{\mathbf{f}} &= \mathbf{0}. \end{aligned} \quad (10)$$

Multiplying the first equation of (10) with the virtual twist rotation $\delta\alpha$ and the second equation of (10) with the virtual displacement vector $\delta\mathbf{r}$, followed by a twofold integration by parts along the beam, results in the following weak form:

$$\begin{aligned} R(\mathbf{r}, \varphi, \delta\mathbf{r}, \delta\alpha) &= \int_0^L \left[\underbrace{\delta\boldsymbol{\theta}^T \mathbf{m}}_{\delta_{rel}\boldsymbol{\omega}^T \mathbf{m}} + \underbrace{\delta\mathbf{r}'^T \check{\mathbf{t}} f_\parallel}_{\delta\epsilon f_\parallel} - \delta\mathbf{r}^T \tilde{\mathbf{f}} - \delta\boldsymbol{\theta}^T \tilde{\mathbf{m}} \right] ds - \left[\delta\mathbf{r}^T \tilde{\mathbf{f}} + \delta\boldsymbol{\theta}^T \tilde{\mathbf{m}} \right]_{\Gamma_\sigma} = 0, \\ \delta\boldsymbol{\theta} &= \delta\alpha \check{\mathbf{t}} + \frac{\mathbf{r}' \times \delta\mathbf{r}'}{\|\mathbf{r}'\|^2}. \end{aligned} \quad (11)$$

In (11), the constrained rotation vector variation according to (7) has been identified. According to the principle of virtual work, the factors in front of the stress resultants $\mathbf{m}$ and $f_\parallel$ in the weak form have to represent the (objective) variation of the corresponding deformation measures, denoted as $\boldsymbol{\omega}$ and ϵ . In this context, the “relative variation” of an arbitrary vector $\mathbf{a} \in \mathfrak{R}^3$ simply denotes the objective part of the variation, which is defined as $\delta_{rel}\mathbf{a} := \delta\mathbf{a} - \delta\boldsymbol{\theta} \times \mathbf{a}$ (see e.g. [4]). One can verify that the deformation measures

$$\boldsymbol{\omega} = \begin{pmatrix} \tau + \varphi' - \tau_0 - \varphi'_0 \\ (\mathbf{g}_2 \cdot \boldsymbol{\kappa}) - (\mathbf{g}_{02} \cdot \boldsymbol{\kappa}_0) \\ (\mathbf{g}_3 \cdot \boldsymbol{\kappa}) - (\mathbf{g}_{03} \cdot \boldsymbol{\kappa}_0) \end{pmatrix}_{\mathbf{g}_i} \quad \text{and} \quad \epsilon := \|\mathbf{r}'\| - 1 \quad (12)$$

are the sought-after antiderivatives of $\delta_{rel}\boldsymbol{\omega}$ and $\delta\epsilon$ fulfilling the initial conditions $\boldsymbol{\omega}(\mathbf{r}_0, \mathbf{A}_0) = \mathbf{0}$ and $\epsilon(\mathbf{r}_0, \mathbf{A}_0) = 0$. Throughout this contribution, the subscript of a matrix denotes the basis (e.g. $\mathbf{g}_i$) in which the corresponding tensor is

Table 1

Problem description for Kirchhoff beams based on arbitrary reference triad definitions $\mathbf{g}_{ref,2}(\mathbf{r})$ and $\mathbf{g}_{ref,3}(\mathbf{r})$.

Weak Form:

$$R(\mathbf{r}, \varphi, \delta\mathbf{r}, \delta\alpha) = \int_0^L \left[\delta\boldsymbol{\theta}^T \mathbf{m} + \delta\mathbf{r}^T \dot{\mathbf{t}} f_{\parallel} - \delta\mathbf{r}^T \tilde{\mathbf{f}} - \delta\boldsymbol{\theta}^T \tilde{\mathbf{m}} \right] ds - \left[\delta\mathbf{r}^T \tilde{\mathbf{f}} + \delta\boldsymbol{\theta}^T \tilde{\mathbf{m}} \right]_{\Gamma_{\sigma}} = 0, \quad \delta\boldsymbol{\theta} = \delta\alpha \check{\mathbf{t}} + \frac{\mathbf{r}' \times \delta\mathbf{r}'}{\|\mathbf{r}'\|^2}.$$

Constitutive law:

$$\mathbf{m} = \mathbf{diag}[GI_T, EI_2, EI_3]_{\mathbf{g}_i} \cdot \boldsymbol{\omega}, \quad f_{\parallel} = EA\epsilon$$

Kinematic equations:

$$\boldsymbol{\omega} = \left(\tau + \varphi' - \tau_0 - \varphi'_0, (\mathbf{g}_2^T \boldsymbol{\kappa}) - (\mathbf{g}_{02}^T \boldsymbol{\kappa}_0), (\mathbf{g}_3^T \boldsymbol{\kappa}) - (\mathbf{g}_{03}^T \boldsymbol{\kappa}_0) \right)_{\mathbf{g}_i}^T, \quad \epsilon = \|\mathbf{r}'\| - 1$$

$$\mathbf{g}_1 = \frac{\mathbf{r}'}{\|\mathbf{r}'\|}, \quad \mathbf{g}_2 = \mathbf{g}_{ref,2} \cos \varphi + \mathbf{g}_{ref,3} \sin \varphi, \quad \mathbf{g}_3 = \mathbf{g}_{ref,3} \cos \varphi - \mathbf{g}_{ref,2} \sin \varphi, \quad \boldsymbol{\kappa} = \frac{\mathbf{r}' \times \mathbf{r}''}{\|\mathbf{r}'\|^2}, \quad \tau = \mathbf{g}_{ref,3}^T \mathbf{g}'_{ref,2}.$$

Boundary conditions:

$$\mathbf{r} = \bar{\mathbf{r}}, \quad \dot{\mathbf{t}} = \check{\mathbf{t}}, \quad \alpha = \bar{\alpha} \text{ on } \Gamma_u \text{ and } \mathbf{f} = \bar{\mathbf{f}}, \quad \mathbf{m} = \bar{\mathbf{m}} \text{ on } \Gamma_{\sigma} \text{ with } \Gamma_{\sigma} \cap \Gamma_u = \emptyset, \quad \Gamma_{\sigma} \cup \Gamma_u = \{0, L\}.$$

represented. The abbreviation $\tau := \mathbf{g}'_{ref,2} \cdot \mathbf{g}_{ref,3} = -\mathbf{g}'_{ref,3} \cdot \mathbf{g}_{ref,2}$ describes the torsion of the reference triad. The last quantity appearing in (12), the vector $\boldsymbol{\kappa}$, is strongly related to the so-called Frenet–Serret curvature and is defined as:

$$\boldsymbol{\kappa} := \check{\mathbf{t}} \times \dot{\mathbf{t}} = \frac{\mathbf{r}' \times \mathbf{r}''}{\|\mathbf{r}'\|^2}. \quad (13)$$

The last ingredient needed to complete the problem description is a constitutive law relating the deformation measures (12) with the corresponding stress resultants $\mathbf{m}$ and $f_{\parallel}$. Throughout this work, we assume a hyperelastic material behavior reflected by the following constitutive relations:

$$f_{\parallel} = EA\epsilon, \quad \mathbf{m} = \mathbf{c}_m \boldsymbol{\omega} \quad \text{with } \mathbf{c}_m = \mathbf{diag}[GI_T, EI_2, EI_3]_{\mathbf{g}_i}. \quad (14)$$

Here, $\mathbf{diag}[GI_T, EI_2, EI_3]$ represents a diagonal matrix with the entries GI_T , EI_2 and EI_3 . In (14), E and G are Young's modulus and the shear modulus, A , $\bar{A}_2$ and $\bar{A}_3$ are the cross-section and the two reduced cross-sections, I_2 and I_3 are the two principal moments of inertia and I_T is the torsional moment of inertia.

Remark. With the beam kinematics presented above the proposed formulation is capable of describing the orientation of cross-sections of any shape. In particular, it is not limited to the simple case of isotropic shapes (such as circular or quadratic cross-sections), often denoted as kinetic symmetry, as it is the case for many Kirchhoff beam formulations that can be found in the literature (see e.g. [35,36] or [37]). However, with the introduction of the specific constitutive law (14), we will from now on confine ourselves to cross-sections whose center of shear coincides with the cross-section centroid (e.g. bi-symmetric cross-sections). While the general beam kinematics presented in Section 2.1 remain valid for each kind of cross-section, some adaptations of the section constitutive law would be necessary if we wanted to drop this confinement.

By introducing the trial space $(\mathbf{r}, \varphi) \in \mathcal{S} \subset \mathfrak{R}^3 \times \mathfrak{R}$ and the weighting space $(\delta\mathbf{r}, \delta\alpha) \in \mathcal{V} \subset \mathfrak{R}^3 \times \mathfrak{R}$, the considered problem is completely defined. The final spatial problem setting including the natural boundary conditions on Γ_{σ} and the essential boundary conditions on Γ_u is summarized in Table 1. It should be emphasized that only the concrete analytical expressions for $\mathbf{g}_{ref,2}(s) = \mathbf{f}_2(\mathbf{r}(s), \mathbf{r}'(s), \mathbf{r}''(s))$ and $\mathbf{g}_{ref,3}(s) = \mathbf{f}_3(\mathbf{r}(s), \mathbf{r}'(s), \mathbf{r}''(s))$ of Table 1 depend on the specific reference triad definition. The reference triad applied within this work will be presented in the next section.

2.3. Definition of the reference triad field

For the derivation of a reference triad field, some basic concepts of spatial discretization are necessary. Thereto, we want to sub-divide the beam into n_{ele} two-noded elements and introduce the elementwise parameter coordinate $\xi \in [-1; 1]$ with nodal values $\xi^1 = -1$ and $\xi^2 = 1$. Throughout this contribution, an upper index refers to the node at which a quantity is evaluated. Furthermore, the Jacobi factor $\|\mathbf{r}_{0,\xi}(\xi)\|$, with $(\cdot)_{,\xi} = d(\cdot)/d\xi$, maps the differential parameter $d\xi$ onto the differential arc-length ds according to $ds = \|\mathbf{r}_{0,\xi}(\xi)\|d\xi$. More detailed information about the spatial discretization will be presented in Section 4. Furthermore, we assume that the external loads are

applied within n_{step} load steps leading to n_{step} static equilibrium solutions at pseudo time values $t_i \in [0; 1]$ with $i = 1, \dots, n_{step}$. Within this work, only the quasi-static analysis of beam problems is considered. However, the construction of a reference triad field as presented below is identical for static and dynamic simulations. For that reason, the terms “time step” and “load step” as well as “time” and “pseudo time” t_i are used as equivalents throughout this work.

The design of an elementwise reference triad field is carried out in two steps: First, we want to define the reference triads $\mathbf{g}_{ref,i}^j(\xi^j)$, with $j = 1, 2$ and $i = 1, 2, 3$, at the nodes. In a second step, these nodal reference triads will be interpolated in the element interior. The nodal reference triads are defined via the “smallest rotation” (SR) mapping (see e.g. [46]) from the nodal tangent $\mathbf{g}_{ref,1,old}^j = \check{\mathbf{t}}_{old}^j$ of the last time step onto the current nodal tangent $\check{\mathbf{t}}^j$:

$$\mathbf{g}_{ref,2}^j := \mathbf{g}_{ref,2,old}^j - \frac{\mathbf{g}_{ref,2,old}^{jT} \check{\mathbf{t}}^j}{1 + \check{\mathbf{t}}_{old}^{jT} \check{\mathbf{t}}^j} (\check{\mathbf{t}}^j + \check{\mathbf{t}}_{old}^j), \quad \mathbf{g}_{ref,3}^j := \mathbf{g}_{ref,3,old}^j - \frac{\mathbf{g}_{ref,3,old}^{jT} \check{\mathbf{t}}^j}{1 + \check{\mathbf{t}}_{old}^{jT} \check{\mathbf{t}}^j} (\check{\mathbf{t}}^j + \check{\mathbf{t}}_{old}^j). \quad (15)$$

Here and in the following, the index *old* refers to a quantity of the last time step, i.e. $(\cdot)_{old} = (\cdot)_{(t_{i-1})}$, while all other terms are evaluated at the current time step t_i . In order to interpolate these nodal reference triads into the element interior, we again apply the SR mapping, but this time from the first nodal reference triad $\mathbf{g}_{ref,1}^1$ at node 1 onto the tangent vector $\check{\mathbf{t}}(\xi)$ at position $\xi \in [-1; 1]$. This procedure leads to:

$$\mathbf{g}_{int,2}(\xi) := \mathbf{g}_{ref,2}^1 - \frac{\mathbf{g}_{ref,2}^{1T} \check{\mathbf{t}}(\xi)}{1 + \check{\mathbf{t}}^1(\xi)^T \check{\mathbf{t}}(\xi)} (\check{\mathbf{t}}(\xi) + \check{\mathbf{t}}^1), \quad \mathbf{g}_{int,3}(\xi) := \mathbf{g}_{ref,3}^1 - \frac{\mathbf{g}_{ref,3}^{1T} \check{\mathbf{t}}(\xi)}{1 + \check{\mathbf{t}}^1(\xi)^T \check{\mathbf{t}}(\xi)} (\check{\mathbf{t}}(\xi) + \check{\mathbf{t}}^1). \quad (16)$$

The base vectors $\mathbf{g}_{int,2}(\xi)$ and $\mathbf{g}_{int,3}(\xi)$ only represent an intermediate triad field. The latter has to be rotated with respect to the tangent vector by the scalar-valued rotation angle field

$$\theta(\xi) = \frac{1 + \xi}{2} \theta^2 \quad \text{with } \theta^2 = \text{sgn}(\mathbf{g}_{int,3}^{2T} \mathbf{g}_{ref,2}^2) \cdot \arccos(\mathbf{g}_{int,2}^{2T} \mathbf{g}_{ref,2}^2), \quad (17)$$

in order to close the gap $\theta^2 \in [-\pi, \pi]$ between the intermediate triad $\mathbf{g}_{int,i}(\xi^2)$ according to (16) and the reference triad $\mathbf{g}_{ref,i}^2$ according to (15) at node 2 and end up with a C^0 -continuous reference triad field $\mathbf{g}_{ref,i}(\xi)$. With the help of (5), (15), (16) and (17), the base vectors $\mathbf{g}_i(\xi)$ can now be expressed exclusively by means of primary variables:

$$\begin{aligned} \mathbf{g}_2(\xi) &= \mathbf{g}_{int,2}(\xi) \cos(\theta(\xi) + \varphi(\xi)) + \mathbf{g}_{int,3}(\xi) \sin(\theta(\xi) + \varphi(\xi)), \\ \mathbf{g}_3(\xi) &= \mathbf{g}_{int,3}(\xi) \cos(\theta(\xi) + \varphi(\xi)) - \mathbf{g}_{int,2}(\xi) \sin(\theta(\xi) + \varphi(\xi)). \end{aligned} \quad (18)$$

Finally, also the torsion of the reference triad field resulting from this procedure has to be expressed in terms of primary variables. It can be shown that the torsion consists of the following two contributions

$$\tau(\xi) = \tau_{int}(\xi) + \theta'(\xi) = \mathbf{g}_{int,2}'(\xi) \cdot \mathbf{g}_{int,3}(\xi) + \frac{d\theta(\xi)}{d\xi} \frac{d\xi}{ds} = -\frac{\kappa(\xi)^T \check{\mathbf{t}}^1}{1 + \check{\mathbf{t}}^1(\xi)^T \check{\mathbf{t}}(\xi)} + \frac{\theta^2}{2\|\mathbf{r}_{0,\xi}(\xi)\|}, \quad (19)$$

where $\tau_{int}(\xi)$ denotes the torsion of the intermediate triad field $\mathbf{g}_{int,i}(\xi)$. This procedure, applied to define the material triad interpolation, will be denoted as *Nodal Smallest Rotation Smallest Rotation Interpolation* (NSRISR) strategy in the following. One can prove that this interpolation strategy leads to an objective beam element formulation, whose spatially discretized deformation measures remain invariant under rigid body motions.

Remark. In [38], it has been stated that the torsion of a triad field defined via the SR mapping according to (16) would vanish. Obviously, this statement cannot be correct, since the torsion presented in (19) only becomes zero in special cases, e.g. for plane curve segments. Via a finite difference scheme, we have verified numerically that the term $\tau_{int}(\xi)$ according to (19) indeed represents the correct torsion of the triad field $\mathbf{g}_{int,i}(\xi)$. Furthermore, it turned out that neglecting the torsion $\tau_{int}(\xi)$ in the weak form of the element formulation would lead to a decrease in the convergence order of the L^2 -error of the primary variable $\mathbf{r}(\xi)$.

2.4. Analytical treatment of Kirchhoff beams

In the last sections, the numerical solution of Kirchhoff beam problems by means of the finite element method has been considered. Within this interlude, we want to show briefly how the concept of reference triads can be beneficial to the analytical treatment of shear-free beams. Thereto, we choose the Frenet–Serret (FS) frame of the curve $\mathbf{r}(s)$, consisting of the tangent vector $\mathbf{t}$, the normal vector $\mathbf{n}_{FS}$ and the binormal vector $\mathbf{b}_{FS}$, as the reference triad field appearing in the problem summary in Table 1. These base vectors are defined as follows:

$$\mathbf{g}_{ref,2} = \mathbf{n}_{FS} := \frac{\|\mathbf{r}'\|}{\|\mathbf{r}' \times \mathbf{r}''\|} \left(\mathbf{r}'' - \|\mathbf{r}'\|^2 (\mathbf{r}'^T \mathbf{r}'') \mathbf{r}' \right) \quad \text{and} \quad \mathbf{g}_{ref,3} = \mathbf{b}_{FS} := \frac{\mathbf{r}' \times \mathbf{r}''}{\|\mathbf{r}' \times \mathbf{r}''\|}. \quad (20)$$

From Eq. (20), it gets obvious that the binormal vector $\mathbf{b}_{FS}$ is parallel to the Frenet–Serret curvature vector

$$\kappa = \kappa \mathbf{g}_{ref,3} \quad \text{with} \quad \kappa = \|\kappa\| = \frac{\|\mathbf{r}' \times \mathbf{r}''\|}{\|\mathbf{r}'\|^2}. \quad (21)$$

After calculating the derivative $\mathbf{g}'_{ref,2}$, we obtain for the torsion $\tau = \mathbf{g}_{ref,3}^T \mathbf{g}'_{ref,2}$ of the Frenet–Serret reference triad:

$$\tau = \frac{\mathbf{r}'^T (\mathbf{r}'' \times \mathbf{r}''') \|\mathbf{r}'\|}{\|\mathbf{r}' \times \mathbf{r}''\|^2}. \quad (22)$$

From a numerical point of view, the Frenet–Serret frame is unfeasible for finite element simulations, since the corresponding base vectors are not defined for curve segments with vanishing curvature, i.e. $\mathbf{r}'' \equiv \mathbf{0}$ (see (20)). Nevertheless, due to its natural adaption to the curve's curvature, the Frenet–Serret frame seems to be an ideal tool for the analytical investigation of Kirchhoff beam problems. For that reason, we want to apply this reference triad and insert the resulting stress resultants into the strong form of the balance Eqs. (10). In the following, we formulate the components of this strong form with respect to the Frenet–Serret frame. Thus, the moment stress resultant, the external force vector and the external moment vector have to be split into corresponding components:

$$\mathbf{m} =: m_{\parallel} \mathbf{t} + m_n \mathbf{n} + m_b \mathbf{b}, \quad \tilde{\mathbf{m}} =: \tilde{m}_{\parallel} \mathbf{t} + \tilde{m}_n \mathbf{n} + \tilde{m}_b \mathbf{b}, \quad \text{and} \quad \tilde{\mathbf{f}} =: \tilde{f}_{\parallel} \mathbf{t} + \tilde{f}_n \mathbf{n} + \tilde{f}_b \mathbf{b}. \quad (23)$$

By projecting the vector-valued equation of (10) into the directions $\mathbf{t}$, $\mathbf{n}$ and $\mathbf{b}$ and using the scalar equation of (10) directly, we get the following set of four differential equations:

$$\begin{aligned} f'_{\parallel} + \frac{\kappa}{1+\epsilon} (\tau m_n + m'_b + \tilde{m}_b) + \tilde{f}_{\parallel} &= 0, \\ - \left(\frac{\tau m_n + m'_b + \tilde{m}_b}{1+\epsilon} \right)' - \frac{\tau}{1+\epsilon} (\kappa m_{\parallel} + m'_n - \tau m_b + \tilde{m}_n) + \kappa f_{\parallel} + \tilde{f}_n &= 0, \\ \left(\frac{-\tau m_b + m'_n + \kappa m_{\parallel} + \tilde{m}_n}{1+\epsilon} \right)' - \frac{\tau}{1+\epsilon} (\tau m_n + m'_b + \tilde{m}_b) + \tilde{f}_b &= 0, \\ m'_{\parallel} - \kappa m_n + \tilde{m}_{\parallel} &= 0. \end{aligned} \quad (24)$$

Furthermore, we get the following expressions for the components of the stress resultants $f_{\parallel}$, $m_{\parallel}$, m_n and m_b :

$$\begin{aligned} f_{\parallel} &= EA\epsilon, \\ m_{\parallel} &= GI_T (\tau + \varphi' - \tau_0 + \varphi'_0), \\ m_n &= EI_2 (\kappa \sin \varphi - \kappa_0 \sin \varphi_0) \cos \varphi - EI_3 (\kappa \cos \varphi - \kappa_0 \cos \varphi_0) \sin \varphi, \\ m_b &= EI_2 (\kappa \sin \varphi - \kappa_0 \sin \varphi_0) \sin \varphi + EI_3 (\kappa \cos \varphi - \kappa_0 \cos \varphi_0) \cos \varphi. \end{aligned} \quad (25)$$

In the case of quasi-circular cross-sections, i.e. $EI_2 = EI_3 =: EI$, the initial material frame $\mathbf{g}_{0i}$ can be chosen as coinciding with the initial reference frame $\mathbf{g}_{ref,0,i}$, thus $\varphi_0 \equiv 0$, which simplifies the bending moment components to $m_n = EI\kappa_0 \sin \varphi$ and $m_b = EI(\kappa - \kappa_0 \cos \varphi)$. Eqs. (25) inserted into Eqs. (24) give a system of four differential equations, which determine the four unknowns ϵ , κ , τ and φ . With the definitions of ϵ , κ and τ as well as appropriate boundary conditions, the curve $\mathbf{r}$ can be determined afterwards. In Section 6, these differential equations will be used to derive analytical solutions for some numerical examples.

3. Reduced beam models

The theory derived in Section 2 has been formulated as general as possible and accounts for beams with non-isotropic cross-section shape and arbitrary initial curvature. In the following, some special cases concerning the beam geometry and the external loads are presented, which enable the application of simplified versions (denoted as reduced models) of the general theory presented in the last section. The motivation for the development of such reduced formulations is twofold: On the one hand, the numerical effort as well as the complexity of the formulation can be reduced drastically. As will be shown in the following, for none of these reduced models any reference or material triad has to be calculated. On the other hand, the special cases presented below are of high practical relevance.

3.1. Two-dimensional problems

In this section, we want to consider beam problems that are restricted to the global $\mathbf{e}_1 - \mathbf{e}_2$ -plane. In this case, we can define without loss of generality a material triad field coinciding with the reference triad field according to:

$$\mathbf{g}_1(s) = \mathbf{g}_{1,ref}(s) = \check{\mathbf{t}}(s), \quad \mathbf{g}_3 = \mathbf{g}_{3,ref} = \mathbf{e}_3 \quad \text{and} \quad \mathbf{g}_2(s) = \mathbf{g}_{2,ref}(s) = \mathbf{e}_3 \times \check{\mathbf{t}}(s). \quad (26)$$

As a consequence of (26), the mechanical torsion as well as the relative angle $\varphi(s)$ vanishes. Furthermore, the Frenet–Serret curvature vector κ remains perpendicular to the global $\mathbf{e}_1 - \mathbf{e}_2$ -plane, which leads to the following weak form:

$$\begin{aligned} R(\mathbf{r}, \delta \mathbf{r}) = & \int_0^L \left[EI_3 \left(\frac{\mathbf{r}' \times \delta \mathbf{r}'}{\|\mathbf{r}'\|^2} \right)^T \left(\frac{\mathbf{r}' \times \mathbf{r}''}{\|\mathbf{r}'\|^2} - \mathbf{r}'_0 \times \mathbf{r}''_0 \right) \right. \\ & \left. + EA \frac{\delta \mathbf{r}'^T \mathbf{r}'}{\|\mathbf{r}'\|} (\|\mathbf{r}'\| - 1) - \delta \mathbf{r}^T \tilde{\mathbf{f}} - \delta \theta \tilde{m} \right] ds - \left[\delta \mathbf{r}^T \tilde{\mathbf{f}} + \delta \theta \tilde{m} \right]_{\Gamma_\sigma} = 0. \end{aligned} \quad (27)$$

In (27), the quantities $\tilde{m}$ and $\tilde{m}$ denote the external distributed and discrete moments pointing in global $\mathbf{e}_3$ -direction. The rotation angle variation $\delta \theta$ also points in global $\mathbf{e}_3$ -direction and has the following analytical representation:

$$\delta \theta = \frac{\delta \mathbf{r}'^T (\mathbf{e}_3 \times \mathbf{r}')}{\|\mathbf{r}'\|^2}. \quad (28)$$

Eqs. (27) and (28) completely define two-dimensional Kirchhoff beams with respect to the primary variable $\mathbf{r}$. As already mentioned above, no reference or material triad field has to be calculated in the presented 2D case.

3.2. Symmetric problems

The case of beams with quasi-circular cross-sections and vanishing initial curvature in a three-dimensional problem setting is denoted as “isotropic bending”-case in the following and characterized by:

$$I_2 = I_3 =: I \quad \text{and} \quad \kappa_0 \equiv 0. \quad (29)$$

3.2.1. Isotropic bending

In the “isotropic bending”-case, the initial configuration of the beam is rotationally symmetric with respect to the centerline. Since any axis lying in the cross-section can be identified as principal axis of inertia, we can choose the initial triad $\mathbf{g}_{0i}(s)$ arbitrarily. For simplicity, we choose it to be constant, i.e. $\mathbf{g}_{0i}(s) \equiv \mathbf{g}_{0i} = \text{const.}$, which implies a vanishing initial torsion $\tau_0 + \varphi'_0 \equiv 0$. By applying the reference triad of Section 2.3, we furthermore let the initial reference triad coincide with the triad $\mathbf{g}_{0i}$. Under these circumstances, the following relations are valid:

$$\tau_0 \equiv 0, \quad \varphi_0 \equiv 0, \quad \kappa_0 \equiv 0, \quad \text{and} \quad I_2 = I_3 = I. \quad (30)$$

Inserting these relations into the corresponding equations of Table 1 leads to the following moment stress resultant:

$$\mathbf{m} = GI_T (\tau + \varphi') \check{\mathbf{t}} + EI \kappa. \quad (31)$$

Since the relative angle φ does not appear in the bending moment components anymore, the substitution $\alpha' := \tau + \varphi'$ can be made, where α' now represents the total torsion. Thus, the primary variable φ is replaced by an alternative degree of freedom α . For that reason, the quantity τ does not have to be calculated and *the resulting weak form does not depend on any reference triad anymore*. While φ describes the orientation of the triad $\mathbf{g}_i$ by means of a relative rotation with respect to a reference triad, the angle α only defines the orientation difference between two triads at s_1 and s_2 in the sense of an accumulated torsion $\Delta\alpha_{12} = \int_{s_1}^{s_2} \alpha' ds$. Inserting these simplifications in the weak form yields:

$$R(\mathbf{r}, \alpha, \delta\mathbf{r}, \delta\alpha) = \int_0^l \left[\delta\mathbf{r}^T \tilde{\mathbf{f}} - \delta\mathbf{r}^T \left(E A \mathbf{h}_1 + E I \mathbf{h}_2 + G I_T \mathbf{h}_3 + \frac{\mathbf{r}'}{\|\mathbf{r}'\|^2} \times \tilde{\mathbf{m}} \right) - \delta\mathbf{r}''^T E I \mathbf{h}_4 \right. \\ \left. + \delta\alpha \frac{\mathbf{r}'^T \tilde{\mathbf{m}}}{\|\mathbf{r}'\|} - \delta\alpha' G I_T \alpha' \right] ds + \left[\delta\mathbf{r}^T \tilde{\mathbf{f}} - \delta\mathbf{r}^T \left(\frac{\mathbf{r}'}{\|\mathbf{r}'\|^2} \times \tilde{\mathbf{m}} \right) + \delta\alpha \frac{\mathbf{r}'^T \tilde{\mathbf{m}}}{\|\mathbf{r}'\|} \right]_{\Gamma_\sigma} = 0. \quad (32)$$

In this weak form, all dependencies on the primary variables $\mathbf{r}$ and α have already been inserted. Furthermore, we have used the following abbreviations:

$$\mathbf{h}_1 := \mathbf{r}' \left(1 - \frac{1}{\|\mathbf{r}'\|} \right), \quad \mathbf{h}_2 := \frac{2\mathbf{r}'(\mathbf{r}'^T \mathbf{r}'')^2}{\|\mathbf{r}'\|^6} - \frac{\mathbf{r}'(\mathbf{r}''^T \mathbf{r}'') + \mathbf{r}''(\mathbf{r}'^T \mathbf{r}'')}{\|\mathbf{r}'\|^4}, \\ \mathbf{h}_3 := \alpha' \frac{\mathbf{r}' \times \mathbf{r}''}{\|\mathbf{r}'\|^3}, \quad \mathbf{h}_4 := \frac{\mathbf{r}''}{\|\mathbf{r}'\|^2} - \frac{\mathbf{r}'(\mathbf{r}'^T \mathbf{r}'')}{\|\mathbf{r}'\|^4}. \quad (33)$$

It should be mentioned that the fields $\alpha(s)$, representing the torsional deformation, and $\mathbf{r}(s)$, representing the bending deformation, are only coupled by one term, namely $\mathbf{h}_3 = \alpha' \frac{\mathbf{r}' \times \mathbf{r}''}{\|\mathbf{r}'\|^3} = \frac{\alpha'}{\|\mathbf{r}'\|} \boldsymbol{\kappa}$, which vanishes in the straight configuration. The weak form (32) is identical to the weak form proposed in [35] for straight Kirchhoff beams with circular cross-sections. Thus, the beam element presented in [35] can be considered a special case that can be derived in a consistent manner from the general formulation given in Table 1.

Remark. Since only the first derivative $\alpha'(s)$ appears in the weak form (32), one could also chose $k_1(s) := \alpha'(s)$ instead of $\alpha(s)$ as primary variable.

3.2.2. Torsion-free applications

In this section, we consider the isotropic bending case of the section above and furthermore assume that there are no torsional components of distributed and discrete external moments acting on the beam, viz.

$$\check{\mathbf{t}}(s) \cdot \tilde{\mathbf{m}}(s) = \tilde{m}_{\parallel}(s) \equiv 0 \quad \text{and} \quad [\check{\mathbf{t}} \cdot \tilde{\mathbf{m}}]_{\Gamma_\sigma} = [\tilde{m}_{\parallel}]_{\Gamma_\sigma} = 0. \quad (34)$$

There are many fields of application where these restrictions may be valid. As examples, we refer to the Brownian dynamics of Actin filaments in biopolymer networks (see e.g. [40]), cables of high voltage lines (see e.g. [47]) or the fibers in high-tensile webbings (see e.g. [42]). Having a closer look at (24) and (25), we realize that the identity $m_{\parallel} \equiv 0$ is valid in the isotropic bending case. Together with the first restriction of (34), the fourth line of (24) becomes

$$m'_{\parallel} = 0. \quad (35)$$

Thus, we have a constant torsional component $m_{\parallel}$ of the moment stress resultant $\mathbf{m}$. Since the torsional components at the beam ends are zero due to the second restriction of (34), the constant value of the torsional component $m_{\parallel}$ is zero along the whole beam, which justifies the notion “torsion-free applications”. On the other hand, there exist applications, e.g. ropes with finite bending stiffness, where the torsional stiffness is negligible, i.e. $G I_T \approx 0$. In common beam theories, such very low torsional stiffnesses would lead to ill-conditioned stiffness matrices. This problem can be sidestepped by completely resigning the degrees of freedom representing the torsional deformation modes. In both cases, the term “torsion-free” is equivalent to “vanishing torsional deformation” and “vanishing torsional stress resultants”, which is in strong contrast to the notion “shear-free” as defined in the introduction. Furthermore, it should again be emphasized that the torsion-free formulation presented in the following is “exact”,

i.e. consistent with the Kirchhoff theory, as long as the restrictions (34) are fulfilled. By removing the torsional contributions from (32), we end up with the following weak form of the torsion-free reduced beam model:

$$R(\mathbf{r}, \delta \mathbf{r}) = \int_0^l \left[\delta \mathbf{r}^T \tilde{\mathbf{f}} - \delta \mathbf{r}^T \left(E A \mathbf{h}_1 + E I \mathbf{h}_2 + \frac{\mathbf{r}'}{\|\mathbf{r}'\|^2} \times \tilde{\mathbf{m}} \right) - \delta \mathbf{r}'^T E I \mathbf{h}_4 \right] ds + \left[\delta \mathbf{r}^T \tilde{\mathbf{f}} - \delta \mathbf{r}^T \left(\frac{\mathbf{r}'}{\|\mathbf{r}'\|^2} \times \tilde{\mathbf{m}} \right) \right]_{\Gamma_\sigma} = 0. \quad (36)$$

The formulation presented in this section is the shear-free counterpart to the torsion-free beam formulation presented in [45], which is based on the Simo–Reissner beam theory. Since the torsion term cancels out from the weak form, only the primary variable $\mathbf{r}$ has to be discretized. The corresponding finite element formulation is based on less degrees of freedom than the general formulation of Section 2.2 and is therefore numerically more efficient. Furthermore, this formulation does not require any rotational degrees of freedom (or any other triad parametrization). This property dramatically simplifies standard procedures such as spatial discretization (in a manner that preserves objectivity of deformation measures), linearization, configuration updates, or time integration via finite differences in dynamics. These procedures are typically complicated by the presence of rotational degrees of freedom in common geometrically exact beam formulations. This reduced beam model also seems to provide an ideal tool for the mechanical investigation of cable-like structures. In pure cable formulations, artificial bending terms are often necessary in order to stabilize compressional modes (see e.g. [44]). On the contrary, our torsion-free formulation naturally provides such a stabilization. Furthermore, the corresponding bending terms are mechanically consistent and deliver accurate results according to the Kirchhoff beam theory as long as the restrictions (34) and (29) are fulfilled.

3.3. Small tension

If the derivation of deformation measures and constitutive relations of the geometrically exact beam theory is to be conducted starting from the general theory of 3D continuum mechanics, the assumption of small axial strains, i.e. $\epsilon = \|\mathbf{r}'\| - 1 \ll 1$, usually becomes necessary (see e.g. [1,19] or [48]). In this section, a simplified version of the weak form shall be derived, which is based on the equivalent assumption $\|\mathbf{r}'\| = \epsilon + 1 \approx 1$ in all terms except the axial tension term itself. By differentiating this relation, it follows that $\mathbf{r}'^T \mathbf{r}'' \approx 0$. This procedure is equivalent to a zeroth-order Taylor expansion of the bending term and a first-order Taylor expansion of the axial tension term in $\|\mathbf{r}'\|$ with reference to $\|\mathbf{r}'\| = 1$. Since this simplification is independent from the assumptions of the sections above, it can also be incorporated into the general problem setting of Table 1. However, the most compact reduced formulation results from inserting the small axial tension assumption into the torsion-free weak form of Section 3.2.2. This yields

$$\int_0^l \left[\delta \mathbf{r}^T \tilde{\mathbf{f}} - \delta \mathbf{r}^T \left(E A \mathbf{r}' (\|\mathbf{r}'\| - 1) - \underbrace{E I \|\mathbf{r}'\|^2 \mathbf{r}'}_{=: \mathbf{b}_1} + \mathbf{r}' \times \tilde{\mathbf{m}} \right) - \underbrace{\delta \mathbf{r}'^T E I \mathbf{r}''}_{=: \mathbf{b}_2} \right] ds + \left[\delta \mathbf{r}^T \tilde{\mathbf{f}} - \delta \mathbf{r}^T (\mathbf{r}' \times \tilde{\mathbf{m}}) \right]_{\Gamma_\sigma} = 0. \quad (37)$$

In (37), only the two terms denoted by $\mathbf{b}_1$ and $\mathbf{b}_2$ remain from the bending contribution to the weak form.

Remark. In our simulations, a considerable sensitivity of this formulation concerning the consistency of the applied zeroth-order Taylor expansion of the bending term has been observed. This means that if e.g. only the assumption $\|\mathbf{r}'\| \approx 1$, but not $\mathbf{r}'^T \mathbf{r}'' \approx 0$, was applied to the weak form according to Eq. (36), the result would be an increase in the discretization error level of the corresponding finite element formulation. Furthermore, the importance of the term $\mathbf{b}_1$ shall be emphasized: If we choose a parametrization $\tilde{\mathbf{r}}(s)$, that exactly fulfills the inextensibility constraint $\|\tilde{\mathbf{r}}'(s)\| \equiv 1$, we would gain a variation that would be perpendicular to this vector, i.e. $\delta \tilde{\mathbf{r}}'^T \tilde{\mathbf{r}}' \equiv 0$. However, in our case, we consider unconstrained, arbitrary variations $\delta \mathbf{r}'$ that do not necessarily have to be perpendicular to the tangent vector $\mathbf{r}'$. Thus, the term $\mathbf{b}_1$ will in general not vanish, even for examples with zero axial tension. Nevertheless, when

linearizing the weak form (37) at the straight configuration, the term $\mathbf{b}_1$ indeed vanishes. In this case, we have b_2 as the only remaining bending term, which is typical for the linear Euler–Bernoulli beam.

4. Discretization

In the beam formulations presented in Sections 2 and 3, $\mathbf{r}''$ represents the highest derivative of the primary variables. The interpolation of the centerline $\mathbf{r}(s)$ consequently requires shape functions that fulfill C^1 -continuity at the element boundaries. Having this requirement in mind, we apply Hermite shape functions as trial functions, leading to two-noded elements with the following representation of the initial and deformed centerline curve:

$$\mathbf{r}_h(\xi) = \sum_{i=1}^2 H_d^i(\xi) \mathbf{d}^i + \frac{l_h}{2} \sum_{i=1}^2 H_t^i(\xi) \mathbf{t}^i \quad \text{and} \quad \mathbf{r}_{0,h}(\xi) = \sum_{i=1}^2 H_d^i(\xi) \mathbf{d}_0^i + \frac{l_h}{2} \sum_{i=1}^2 H_t^i(\xi) \mathbf{t}_0^i. \quad (38)$$

In the following, the index $(\cdot)_h$ refers to the discrete representation of a quantity. In (38), the three-dimensional vectors $\mathbf{d}^i$, $\mathbf{d}_0^i$, $\mathbf{t}^i$ and $\mathbf{t}_0^i$ represent nodal positions and nodal tangents, respectively, and again $\xi \in [-1, +1]$ is the normalized element coordinate. Furthermore, the constant l_h denotes the element length on the discretized initial geometry and can be calculated via the iterative solution of the following equation (see e.g. [22]):

$$l_h := \int_{-1}^1 \|\mathbf{r}_{0,h,\xi}(\xi, l_h)\| d\xi. \quad (39)$$

The Hermite shape functions applied in (38) have the following polynomial representation:

$$\begin{aligned} H_d^1(\xi) &= \frac{1}{4}(2 + \xi)(1 - \xi)^2, & H_d^2(\xi) &= \frac{1}{4}(2 - \xi)(1 + \xi)^2, \\ H_t^1(\xi) &= \frac{1}{4}(1 + \xi)(1 - \xi)^2, & H_t^2(\xi) &= -\frac{1}{4}(1 - \xi)(1 + \xi)^2. \end{aligned} \quad (40)$$

These shape functions fulfill the completeness conditions of third-order polynomials as well as the interpolation property with respect to the functional value and the first derivative. If a given initial geometry $\mathbf{r}_0(s)$ is to be discretized by means of (38), one has to define $n_{ele} + 1$ nodes lying on $\mathbf{r}_0(s)$, thus determining n_{ele} elements. It is assumed that either an analytical description of $\mathbf{r}_0(s)$ exists, or at least the nodal positions $\mathbf{r}_0(s^i)$ and tangents $\mathbf{r}'_0(s^i)$, with s^i representing the arc-length coordinates of the nodes, are known. Since s represents the arc-length parameter of the initial geometry, the initial tangent vector field and the initial nodal tangent vectors have to be of unit length, i.e. $\|\mathbf{r}'_0(s)\| \equiv 1 \rightarrow \|\mathbf{r}'_0(s^i)\| = 1$. The initial nodal positions and tangents are then chosen as $\mathbf{d}_0^1 := \mathbf{r}_0(\xi^1)$, $\mathbf{d}_0^2 := \mathbf{r}_0(\xi^2)$, $\mathbf{t}_0^1 := \mathbf{r}'_0(\xi^1)$ and $\mathbf{t}_0^2 := \mathbf{r}'_0(\xi^2)$ for each element. Similarly, due to the interpolation property of the Hermite shape functions, we have the relations $\mathbf{r}(\xi^1) = \mathbf{d}^1$, $\mathbf{r}(\xi^2) = \mathbf{d}^2$, $\mathbf{r}'(\xi^1) = \mathbf{t}^1$ and $\mathbf{r}'(\xi^2) = \mathbf{t}^2$. It should be emphasized that as a consequence of axial tension (see e.g. (12)) and in contrast to the initial nodal tangents $\mathbf{t}_0^i$, the current nodal tangents $\mathbf{t}^i$ appearing in (38) are in general no unit vectors, i.e. $\epsilon(\xi^i) = \|\mathbf{r}'(\xi^i)\| - 1 = \|\mathbf{t}^i\| - 1 \neq 0 \rightarrow \|\mathbf{t}^i\| \neq 1$.

Furthermore, the map between the discrete arc-length increment and an increment of ξ is given by the Jacobi factor $ds_h = \|\mathbf{r}_{0,h,\xi}(\xi)\| d\xi$, which is needed in order to compute derivatives with respect to s_h . For the interpolation of the variable $\varphi(s)$, we apply standard C^0 -continuous Lagrange polynomials $L^i(\xi)$, i.e.

$$\varphi_h(\xi) = \sum_{i=1}^3 L^i(\xi) \varphi^i, \quad (41)$$

with the nodal relative angles φ^i . Since $\varphi(s)$ describes the triad orientation, which scales with $\mathbf{r}'(s)$, a second-order interpolation is sufficient for the relative angle field. All in all, the presented beam element consists of two boundary nodes at $\xi^1 = -1$ and $\xi^2 = 1$ with seven degrees of freedom ($\mathbf{d}^i$, $\mathbf{t}^i$, φ^i) each, and one interior node at $\xi_3 = 0$ with one degree of freedom φ^i . Furthermore, following a Bubnov–Galerkin approach, the discretized variations are

$$\delta \mathbf{r}_h(\xi) = \sum_{i=1}^2 H_d^i(\xi) \delta \mathbf{d}^i + \frac{l_h}{2} \sum_{i=1}^2 H_t^i(\xi) \delta \mathbf{t}^i \quad \text{and} \quad \delta \alpha_h(\xi) = \sum_{i=1}^3 L^i(\xi) \delta \alpha^i. \quad (42)$$

With (38), (41) and (42) the discretized problem setting is completed. All trial and weighting functions have been replaced by their discrete counterparts taken from the finite-dimensional trial subspace $(\mathbf{r}_h, \varphi_h) \in \mathcal{S}_h \subset \mathcal{S}$ and the weighting subspace $(\delta \mathbf{r}_h, \delta \alpha_h) \in \mathcal{V}_h \subset \mathcal{V}$.

Remark. In the case of the “isotropic bending” reduced model of Section 3.2.1, the relative angle $\varphi(s)$ is replaced by the total twist angle $\alpha(s)$ as fourth degree of freedom. Analogously, the discrete representation $\alpha_h(\xi)$ is replaced by an interpolation identical to that of $\varphi_h(\xi)$ according to (41).

5. Minimal Constraint Strains (MCS)

It is well-known that purely displacement-based finite elements are prone to locking. Locking effects particularly relevant for geometrically exact beam formulations are shear locking as well as membrane locking. While shear locking can – by definition – only appear in beam formulations of Simo–Reissner type, membrane locking has already been observed for geometrically linear Kirchhoff beams (see e.g. [22] or [34]). In general, membrane locking refers to the inability of elements to exactly reproduce inextensibility for curved structures (e.g. shells or beams). This is due to a coupling between the kinematic quantities describing the axial tension mode and the curved geometry.

In the case considered here, this kinematic quantity is the discrete centerline curve $\mathbf{r}_h(\xi)$, which influences the axial tension $\epsilon_h(\xi) = \|\mathbf{r}'_h(\xi)\| - 1$ as well as the Frenet–Serret curvature according to (13). In the following, we want to investigate the possibility of exactly reproducing a constant value $\|\mathbf{r}'_h(\xi)\| = 1$, which is equivalent to a vanishing axial tension. For simplicity, we assume an initially straight beam element with length l leading to the constant Jacobi factor $\|\mathbf{r}_{0,h,\xi}(\xi)\| = l/2$. According to Section 4, we obtain for the norm of the derivative of the centerline curve:

$$\begin{aligned} \|\mathbf{r}'_h(\xi)\| &= \frac{2}{l} \left\| \frac{1}{8} \underbrace{(-6\mathbf{d}^1 + 6\mathbf{d}^2 - l\mathbf{t}^1 - l\mathbf{t}^2)}_{\|0\| \doteq 4l} + \frac{\xi}{8} \underbrace{(-2l\mathbf{t}^1 + 2l\mathbf{t}^2)}_{\doteq 0} + \frac{\xi^2}{8} \underbrace{(6\mathbf{d}^1 - 6\mathbf{d}^2 + 3l\mathbf{t}^1 + 3l\mathbf{t}^2)}_{\doteq 0} \right\| \\ &\doteq 1. \end{aligned} \quad (43)$$

The only solution fulfilling the requirements indicated by the curly brackets in order to display $\|\mathbf{r}'_h(\xi)\| = 1$ are

$$\mathbf{t}^2 = \mathbf{t}^1 =: \mathbf{t} \quad \text{and} \quad \mathbf{d}^2 - \mathbf{d}^1 = l\mathbf{t} \quad \text{with} \quad \|\mathbf{t}\| = 1, \quad (44)$$

which leads to the following constant expression for the derivative $\mathbf{r}'_h(\xi)$ of the centerline position vector field:

$$\mathbf{r}'_h(\xi) = \mathbf{t} \quad \text{with} \quad \|\mathbf{t}\| = 1. \quad (45)$$

Thus, the only possible solution with $\|\mathbf{r}'_h(\xi)\| = 1$ is a straight line. In other words, inextensibility cannot be reproduced exactly for curved beam elements, which is just the definition of membrane locking. In order to further investigate the impact of membrane locking on the presented Kirchhoff beam and to develop solution strategies, the following simple example will be investigated: We consider an initially straight beam of length l with square cross-section of side length r ($A = r^2$ and $I_2 = I_3 = r^4/12$), clamped on its left end, loaded with a discrete end-moment at its right end and discretized by one finite element. The external load is chosen as $M = EI_3\pi/(2l)$, which leads to a deformed configuration represented by a quarter circle (see Fig. 2(a)).

Next, we want to investigate the energy contributions of the axial tension and the bending mode. For simplicity, we choose the following nodal values, which represent the nodal position and tangent vectors of the analytical solution with respect to the given coordinate lines x , y and z :

$$\mathbf{d}^1 = (0, 0, 0)^T, \quad \mathbf{d}^2 = l \underbrace{(2/\pi, -2/\pi, 0)^T}_{=: \tilde{\mathbf{d}}_2}, \quad \mathbf{t}^1 = (1, 0, 0)^T, \quad \mathbf{t}^2 = (0, 1, 0)^T. \quad (46)$$

Here, we have introduced the dimensionless position vector $\tilde{\mathbf{d}}_2$ of the right node of the beam. Furthermore, we want to assume the reduced weak form of Section 3.2.2, which delivers exactly the same results as the general formulation of Section 2 for the considered case of an initially straight beam without torsional loads. In this case, the total length-

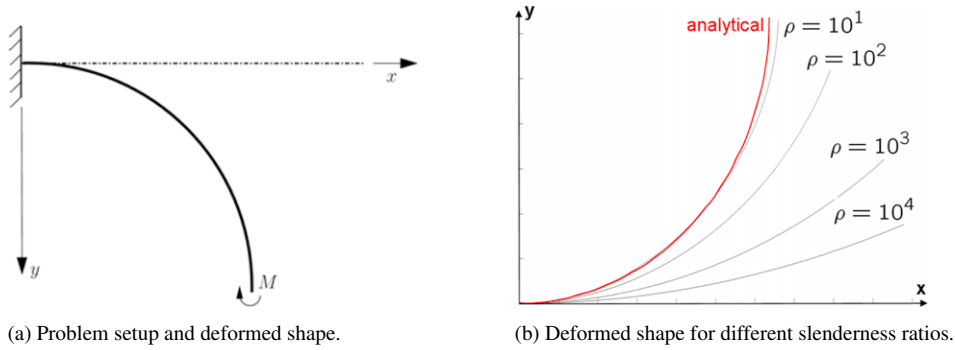


Fig. 2. Initially straight clamped beam bent to a quarter circle by a discrete end-moment $M = EI\pi/(2l)$.

specific elastic energy of the beam reads

$$V_h = V_{th} + V_{bh} = \frac{1}{2}EA\epsilon_h^2 + \frac{1}{2}EI_3\kappa_h^2, \quad (47)$$

where V_{th} and V_{bh} denote the discrete energy contributions of the axial tension and the bending deformation, respectively. By using the result of (43), we obtain for the discretized axial tension:

$$\epsilon_h = \|\mathbf{r}'_h(\xi)\| - 1 = \frac{1}{4} \left\| \left(6\tilde{\mathbf{d}}^2 - \mathbf{t}^1 - \mathbf{t}^2 \right) + 2\xi \left(\mathbf{t}^2 - \mathbf{t}^1 \right) + 3\xi^2 \left(-2\tilde{\mathbf{d}}^2 + \mathbf{t}^1 + \mathbf{t}^2 \right) \right\| - 1 =: \tilde{\epsilon}_h. \quad (48)$$

With the results of Section 4, we can now give an expression for the discretized Frenet–Serret curvature κ_h . For simplicity, it is assumed here that the small tension assumption of Section 3.3 holds. The latter can be considered as a reasonable approximation for this example, where the analytical solution exactly fulfills $\|\mathbf{r}'\|(s) \equiv 1$. This yields:

$$\begin{aligned} \kappa_h &\approx \|\mathbf{r}'_h \times \mathbf{r}''_h\| = \frac{1}{2l} \left\| \left[3\tilde{\mathbf{d}}^2 \times (\mathbf{t}^2 - \mathbf{t}^1) - \mathbf{t}^1 \times \mathbf{t}^2 \right] + \xi \left[6\tilde{\mathbf{d}}^2 \times (\mathbf{t}^2 + \mathbf{t}^1) \right] \right. \\ &\quad \left. + \xi^2 \left[3\tilde{\mathbf{d}}^2 \times (\mathbf{t}^2 - \mathbf{t}^1) - 3\mathbf{t}^1 \times \mathbf{t}^2 \right] \right\| =: \frac{\tilde{\kappa}_h}{l}. \end{aligned} \quad (49)$$

In (48) and (49), we have introduced the corresponding dimensionless quantities $\tilde{\epsilon}_h$ and $\tilde{\kappa}_h$. With these deformation measures at hand, we can formulate the proportion of axial tension energy and bending energy:

$$\frac{V_{th}}{V_{bh}} = 12 \left(\frac{l}{r} \right)^2 \left(\frac{\tilde{\epsilon}_h}{\tilde{\kappa}_h} \right)^2. \quad (50)$$

Since the dimensionless proportion $\tilde{\epsilon}_h/\tilde{\kappa}_h$ is independent of the problem scaling, the proportion of the parasitic axial strain energy to the bending strain energy scales quadratically with the element slenderness ratio $\rho := l/r$. Thus, the key parameter for the considered locking phenomenon is the ratio of the element length l and the “effective cross-section radius” r . This is in agreement with the investigations for geometrically linear Kirchhoff elements in [22].

The simulation results for $E = 1.0$, $l = 1000$ and different slenderness ratios ($\rho = 10$, $\rho = 100$, $\rho = 1000$ and $\rho = 10000$) as well as the analytical solution are presented in Fig. 2(b). Of course, for each slenderness ratio, the external moment has been adapted in order to yield a quarter circle as analytical solution. As expected, an increasing slenderness ratio aggravates the locking effect and leads to a progressively stiff system answer.

In Fig. 3, the numerical results of $\|\mathbf{r}'_h(\xi)\|$ calculated via exact Gauss integration are plotted for simulations with different slenderness ratios. With increasing slenderness ratio, the ratio of axial stiffness to bending stiffness increases, which leads to a progressively strong penalization of the parasitic strains. Correspondingly, the FEM solutions show a decrease in the parasitic strains for increasing slenderness ratios. From Fig. 3, it also gets obvious that the positions of the points fulfilling $\|\mathbf{r}'_h(\xi)\| = 1$ are almost identical for the different slenderness ratios. Since $\mathbf{r}'(\xi)$ is a dimensionless quantity, the evolutions illustrated in Fig. 3 are independent of the problem scaling and valid whenever a quarter circle is discretized by one finite element. However, with finer discretizations, the arc segment represented by one finite element shrinks. One can show, that the fluctuation of $\|\mathbf{r}'(\xi)\|$ decreases in this case.

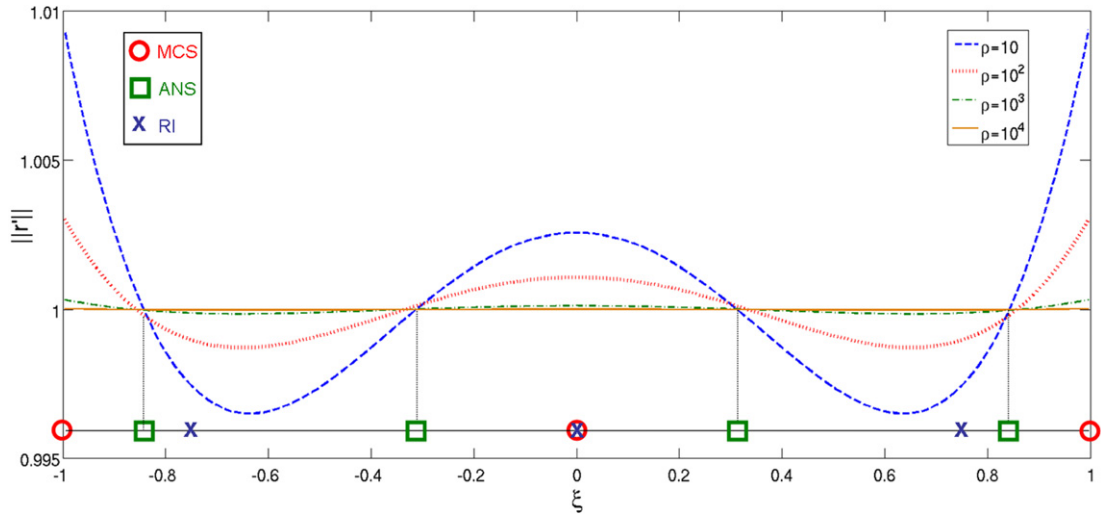


Fig. 3. Initially straight beam bent to a quarter circle by a discrete end-moment: Evolution of $\|\mathbf{r}'(\xi)\|$ within an element. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Remark. Actually, the discretized weak form of Table 1 consists of functional expressions that cannot be integrated exactly by Gauss integration. However, in numerical simulations, the deviations in the results for a number of Gauss points $n_G \geq 4$ turned out to be very small as compared with the discretization error. Therefore, the notion “exact integration” will be used whenever four or more Gauss points have been applied.

Different solution strategies exist in order to get rid of these undesirable locking effects. In the following, we want to compare the approach of Assumed Natural Strains (ANS, see e.g. [49]), Reduced Integration (RI, see e.g. [25]) and a new alternative procedure, which will be denoted as Minimal Constraint Strains (MCS). This new method can be characterized as an assumed strain/B-bar approach (see e.g. [50]) and can therefore be motivated in a variationally consistent manner. Concretely, the axial tension term is replaced by an assumed axial tension term:

$$R_\epsilon = \underbrace{\delta \mathbf{r}'^T \mathbf{t}}_{\delta \epsilon} E A \underbrace{(\|\mathbf{r}'\| - 1)}_{\epsilon} \rightarrow R_{\bar{\epsilon}} = \delta \bar{\epsilon} E A \bar{\epsilon}$$

$$\text{with } \bar{\epsilon}(\xi) = \sum_{k=1}^{n_{CP}} L_k^{n_{CP}}(\xi) \epsilon(\xi_k) \quad \text{and} \quad \delta \bar{\epsilon}(\xi) = \sum_{k=1}^{n_{CP}} L_k^{n_{CP}}(\xi) \delta \epsilon(\xi_k). \quad (51)$$

In (51), ξ_k denotes the parameter coordinate and n_{CP} the number of the Collocation Points (CPs) where the “original” strains are evaluated. Moreover, $L_k^{n_{CP}}(\xi)$ are Lagrange shape functions of polynomial order $n_{CP} - 1$. By linearizing the axial tension term (51) of the weak form at the undeformed, straight configuration, one can identify the B-bar structure typical for geometrically linear finite elements:

$$LIN(R_\epsilon)|_{\mathbf{r}_0} = \delta \mathbf{U}^T \underbrace{\mathbf{H}'^T \mathbf{t}_0}_{\mathbf{B}^T} E A \underbrace{\mathbf{t}_0^T \mathbf{H}'}_{\mathbf{B}} \Delta \mathbf{U} \rightarrow LIN(R_{\bar{\epsilon}})|_{\mathbf{r}_0} = \delta \mathbf{U}^T \bar{\mathbf{B}}^T E A \bar{\mathbf{B}} \Delta \mathbf{U}$$

$$\text{with } \bar{\mathbf{B}}(\xi) = \sum_{k=1}^{n_{CP}} L_k^{n_{CP}}(\xi) \mathbf{B}(\xi_k). \quad (52)$$

Here, the vector $\mathbf{U}$ summarizes all discrete primary variables and the matrix $\mathbf{H}(\xi)$ contains the Hermite shape functions presented in Section 4, such that the displacement field can be calculated as $\mathbf{u}_h(\xi) := \mathbf{r}_h(\xi) - \mathbf{r}_{0,h}(\xi) = \mathbf{H}(\xi) \mathbf{U}$.

Remark. In the authors’ previous work [39], it has been shown that the strain field $\epsilon(\xi)$ is objective and path-independent, meaning that the strain field does not change as consequence of a rigid body motion and that its value at a certain configuration is independent from the deformation path leading to that configuration. Since the assumed

strain field $\bar{\epsilon}(\xi)$ represents a pure re-interpolation of the original strain field $\epsilon(\xi_k)$ evaluated at fixed collocation points ξ_k , the former will also fulfill the properties of objectivity and path-independence.

The essential question now is, how to choose the number and locations of the CPs. As indicated by means of the green squares in Fig. 3, the ANS approach is based on an interpolation of the axial tension values at the CPs with vanishing axial tension. Analogously, the blue crosses indicate the integration points (IPs) of a reduced Gauss integration scheme with three Gauss points. The MCS method, however, applies a third set of CPs, marked with red circles in Fig. 3. In the following, the special choice of these collocation points will briefly be motivated. Thereto, we again want to have a look at the quarter circle example as described above. If we consider the example as a plane problem discretized by one beam element, we have the following eight nodal degrees of freedom:

$$d_1^1, d_2^1, d_1^2, d_2^2, t_1^1, t_2^1, t_1^2, t_2^2. \quad (53)$$

Three of these nodal values are prescribed by the Dirichlet boundary conditions of a clamped end:

$$d_1^1 = 0, \quad d_2^1 = 0, \quad t_1^1 = 0. \quad (54)$$

Here, the subscripts denote the spatial dimensions of the considered 2D problem. Of course, locking could be avoided, if the beam element represented the internal energy according to (47) exactly. If we require the latter for the considered 2D case, three conditions follow from the approximated discretized curvature vector in (49), which demand that the coefficients in front of ξ and ξ^2 vanish and that the norm of the constant part of the curvature vector equals the constant analytical curvature of the quarter circle, i.e. $\kappa = \pi/(2l)$. Thus, from (49), we already have three equations for the five remaining degrees of freedom. We have already observed that it is not possible to fulfill the identity $\epsilon_h \equiv 0$ on the entire element for curved geometries. Thus, the number of collocation points determines the number of positions, where $\epsilon_h(\xi_k) = 0$ has to be fulfilled. If we chose e.g. the two element nodes as CPs, we would get the two additional equations $\|\mathbf{r}'_h(-1)\| \doteq 1$ and $\|\mathbf{r}'_h(1)\| \doteq 1$ in order to end up with the correct axial tension energy.

This way, we would get a system of equations with $n_{eq} = n_{uk}$, where n_{eq} denotes the number of equations and n_{uk} the number of unknowns after boundary conditions have been imposed. It has been verified numerically that this choice of CPs indeed is able to produce FEM solutions $\mathbf{r}_h(\xi)$ that exactly represent the analytical internal energy, i.e. $V_{th} = 0$ and $V_{tb} = \frac{1}{2}EI_3 \frac{\pi^2}{4l^2}$. However, this choice has one crucial drawback: The coefficient equations in (49) do not necessarily have to be independent from each other for arbitrary beam configurations. The consequence can be an ill-conditioned, in extreme cases even a singular, stiffness matrix. To demonstrate this, we want to have a closer look at the straight configuration of the beam element. It is easy to verify that the fulfillment of the two equations

$$d_2^2 \doteq 0 \quad \text{and} \quad t_2^2 \doteq 0 \quad (55)$$

is sufficient in order to reproduce a vanishing curvature along the entire beam element. Thus, in the straight configuration, which can be considered as the most critical case, three additional equations stemming from the axial tension mode are necessary in order to ensure $n_{eq} = n_{uk}$. In this case, the choice of only two CPs would lead to a singular tangent stiffness matrix, a result which can be traced back to the existence of a zero-energy mode in the straight configuration. In order to avoid zero-energy modes of this type and a corresponding rank deficient stiffness matrix, we have to increase the number of CPs to three. Therefore, we choose the element midpoint as third CP.

This procedure can be generalized to discretizations with n_{ele} elements leading to $2 \cdot n_{ele}$ equations of type (55) in order to enforce the straight beam configuration. The choice of CPs according to Fig. 3 delivers $n_{eq,c} = 2 \cdot n_{ele} + 1$ further equations representing the inextensibility constraint. Here and in the following, $n_{eq,c}$ denotes the number of equations due to the axial tension constraints at the CPs. All in all, we have $n_{eq} = 4 \cdot n_{ele} + 1$ equations. After imposing three Dirichlet boundary conditions of the type (54), which is the minimal number for a quasi-static 2D problem, we have $n_{uk} = 4 \cdot (n_{ele} + 1) - 3 = 4 \cdot n_{ele} + 1$ unknowns. Again, the relation $n_{eq} = n_{uk}$ holds true. Thus, this choice leads to the minimal number of CPs necessary in order to avoid a singular stiffness matrix in the critical straight configuration, which explains the naming of the new approach as Minimal Constraint Strains.

The following considerations should further motivate this special choice of CPs: By choosing two CPs in the element interior, we would get $n_{eq,c} = 2 \cdot n_{ele}$ constraint equations and all in all a singular system of equations with $n_{eq} = 4 \cdot n_{ele} < n_{uk} = 4 \cdot n_{ele} + 1$. From this result, we conclude that at least three CPs per element are necessary. For symmetry reasons, the second CP is chosen as the element midpoint. If we now chose the first and third CP in

the element interior, we would get $n_{eq,c} = 3 \cdot n_{ele}$ constraint equations and all in all an over-determined system of equations with $n_{eq} = 5 \cdot n_{ele} > n_{uk} = 4 \cdot n_{ele} + 1$. For that reason, we have chosen the element nodes as first and third CP with each node delivering **one** constraint equation for the **two** elements adjacent at this node. However, this principle only works due to the applied C^1 -continuous Hermite interpolation of the beam centerline leading to C^0 -continuous axial strains $\epsilon(\xi)$ at the nodes. Otherwise, i.e. in case of a C^0 -continuous centerline and C^{-1} -continuous strains, each node would deliver **two** constraint equations for the **two** elements adjacent at this node. In this case, the nodes would have to be considered as interior CPs leading again to $n_{eq,c} = 3 \cdot n_{ele}$ constraint equations and an over-determined system. Of course, for arbitrarily curved configurations, i.e. in cases where the requirements following from (49) are independent from each other, this procedure still leads to a slightly over-determined system of equations. However, the number of constraints and therefore the degree of over-determination is much lower than for the original tension term, which already provides five constraint equations per element for the considered 2D case. Furthermore, these findings suggest that a reduced integration with only two Gauss points or an ANS procedure with only two CPs would also lead to a singular stiffness matrix. Of course, these considerations can be directly extended to the general 3D case.

Finally, we want to consider a further criterion in order to investigate the presented MCS method. As introduced in [51], and for example applied in [52], the so-called constraint ratio allows for some, at least heuristic, evaluation of the locking behavior of a finite element formulation. The constraint ratio is defined as the ratio of the total number of equilibrium equations to the total number of constraint equations:

$$r := \frac{n_{eq}}{n_{eq,c}}. \quad (56)$$

In order to analyze the locking behavior, the constraint ratio r of the continuous problem and the constraint ratio r_h of the discretized problem evaluated for an infinite number of elements have to be compared. The underlying proposition is that elements with $r_h < r$ (especially $r_h < 1$, which means that more constraints than degrees of freedom are present) have a tendency to lock, whereas values $r_h > r$ of the constraint ratio indicate that not enough constraint equations are available in order to reproduce the constraint in an accurate manner. Following this hypothesis, the case $r_h = r$ has to be regarded as the optimal constraint ratio. The 2D space-continuous problem of the considered Kirchhoff beam is described by two differential equations, which determine the two components of the centerline curve $\mathbf{r}(s)$, while we have one constraint equation, namely $\|\mathbf{r}(s)'\| \equiv 1$. Consequently, we get a constraint ratio of $r = 2$ for the continuous problem. The constraint ratio of the spatially discretized 2D problem in the limit of $n_{ele} \rightarrow \infty$ follows as:

$$r_h = \lim_{n_{ele} \rightarrow \infty} \frac{4 \cdot n_{ele} + 1}{2 \cdot n_{ele} + 1} = 2 = r \checkmark. \quad (57)$$

Thus, the choice of CPs indicated in Fig. 3 gives the optimal constraint ratio for the 2D case. Similarly, one can show that this choice also leads to an optimal constraint ratio of $r_h = r = 4$ for the general 3D case. Furthermore, the higher constraint ratio of 3D problems suggests that membrane locking may play a less crucial role than in the 2D case.

All in all, it can be concluded that the choice of the element nodes and the element midpoint as collocation points leads to the minimal possible number of constraint equations $n_{eq,c} = 2 \cdot n_{ele} + 1$. The functional principle of the well-known reduced integration procedure is the reduction of constraint equations, too. However, the minimal possible number of Gauss integration points and therefore of constraint equations $n_{eq,c} = 3 \cdot n_{ele}$ is higher than for the MCS method. The concept of the ANS approach is somehow different: There, the points where the constraint is already fulfilled by the original formulation are used as collocation points. This procedure can avoid the locking effect independent of the number of constraint equations. The drawback of the ANS method is that the position of these points may change for more general deformation states in the geometrically nonlinear regime. Thus, from a theoretical point of view, our new MCS method seems to be the most promising approach in order to avoid membrane locking. The numerical results in Section 6 will confirm this assumption.

6. Numerical results

In this section, we want to present several numerical examples, all of which were conducted with our in-house multiphysics research code BACI. Some of the examples feature special load cases and geometries allowing for the

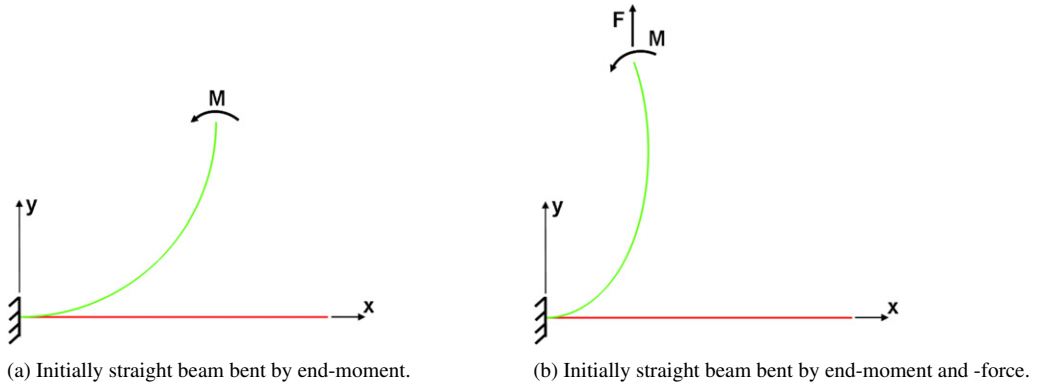


Fig. 4. Problem setup and deformed configuration of an initially straight beam for two different load cases.

application of the reduced models presented in Section 3. Others are characterized by highly slender geometries in order to investigate the locking behavior of the beam formulation and to compare the different methods for locking avoidance presented in Section 5. The standard numerical integration rule applied for the examples within this section is a 4 point Gauss quadrature scheme. Furthermore, a Newton–Raphson scheme is applied in order to solve the nonlinear system of equations $\mathbf{R}_h$ resulting from the discretized version of the weak form according to Table 1. As convergence criteria we check the Euclidean norms of the displacement increment vector $\Delta \mathbf{U}$ and of the residual vector $\mathbf{R}_h$. For convergence, these norms have to go below prescribed tolerances $\delta_{\mathbf{R}_h}$ and $\delta_{\mathbf{U}}$, i.e. $\|\mathbf{R}_h\| < \delta_{\mathbf{R}_h}$ and $\|\mathbf{U}\| < \delta_{\mathbf{U}}$. Since the consistent linearization of the residual terms, which has been carried out using both analytical techniques and automatic differentiation tools, is rather straightforward, it is not presented within this paper.

For simplicity, we have defined a standard set of geometrical and constitutive parameters applied in all simulations unless stated otherwise. This standard set consists of a beam with an initial length $l = 1000$ and a square cross-section with side length r . Different beam slenderness ratios $\rho := l/r$ are generated by varying the cross-section side length. In our simulations the values $r = 100, 10, 1, 0.1$, leading to $\rho = 10, 100, 1000, 10\,000$, are investigated. Furthermore, the constitutive parameters are chosen as $E = 1.0$ and $G = 0.5$, which leads to $EI = GI_T$. Thanks to these standard parameter choices, we can also choose fixed values for the convergence tolerances, viz. $\delta_{\mathbf{U}} = 10^{-8}$ and $\delta_{\mathbf{R}_h} = 10^{-7}, 10^{-9}, 10^{-11}, 10^{-13}$ for the slenderness ratios $\rho = 10, 100, 1000, 10\,000$. In all convergence plots investigated within this section, the following definition of the relative L^2 -error $\|e\|_{rel}^2$ has been employed:

$$\|e\|_{rel}^2 = \frac{1}{u_{max}} \sqrt{\frac{1}{l} \int_0^l \|\mathbf{r}_h - \mathbf{r}_{ref}\|^2 ds}. \quad (58)$$

Herein, $\mathbf{r}_h$ denotes the numerical solution of the beam centerline position for a certain discretization. The normalization with the element length l makes the error independent of the length of the considered beam. The second normalization leads to a more convenient relative error measure, which relates the L^2 -error to the maximal displacement u_{max} occurring for the investigated load case. For all examples without analytical solution, the standard choice for the reference solution $\mathbf{r}_{ref}$ is a numerical solution via the general Kirchhoff beam element presented in Section 2 and denoted as NSRISR element with a spatial discretization that is by a factor of four finer than the finest discretization shown in the corresponding convergence plot.

6.1. Example 1: Membrane locking

An initially straight beam is clamped at one end. For this geometry, two different load cases will be analyzed: The first load case (“M”) is identical to the example of Section 5 and solely consists of a discrete end-moment $\mathbf{M} = (0, 0, M)^T$ applied in one load step. The moment $M = EI\pi/(2l)$ exactly bends the beam into a quarter-circle shaped arc. In the second load case (“M + F”), the end-moment and an additional tip force $\mathbf{F} = (0, F, 0)^T$ in global y-direction are applied in one load step. The initial and deformed geometries for these two load cases are illustrated in Fig. 4.

Table 2

Loadcase M: tip displacement and corresponding relative error for different formulations and meshes.

Formulation	# Elements	$u_x(l)$	$u_y(l)$	$u_z(l)$	$e_{rel}(l)$
Analytical	–	–363.3802276	636.6197724	–	–
NSRISR	1	–351.4109058	631.7879834	–	$1.8 \cdot 10^{-2}$
NSRISR	64	–363.3802266	636.6197719	–	$1.5 \cdot 10^{-9}$

Table 3

Loadcase M + F: tip displacement and corresponding relative error for different formulations and meshes.

Formulation	# Elements	$u_x(l)$	$u_y(l)$	$u_z(l)$	$e_{rel}(l)$
Reissner	8192	–714.2038023	837.2855039	–	–
NSRISR	1	–721.5087997	801.2828142	–	$3.3 \cdot 10^{-2}$
NSRISR	64	–714.2038024	837.2855030	–	$8.0 \cdot 10^{-10}$

While the standard parameters and the different slenderness ratios $\rho = 10, 100, 1000, 10\,000$ have been chosen for the load case M, only the highest and therefore most critical slenderness ratio $\rho = 10\,000$ in combination with an external force of $F = 1.0 \cdot 10^{-10} \approx 10 \cdot M/l$ has been investigated for the load case M + F. We apply the NSRISR element formulation in combination with a full Gauss integration (“FI”) with $n_G \geq 4$, a reduced Gauss integration (“RI”) with $n_G = 3$, a classical assumed natural strain approach (“ANS”) as well as the new minimal constraint strains approach (“MCS”) as presented in Section 5.

In order to get a first idea of the resulting deformation and the accuracy of the formulation, the tip displacements $\mathbf{u}(l)$ and the corresponding relative errors $e_{rel}(l) := \|\mathbf{u}(l) - \mathbf{u}_{ref}(l)\| / \|\mathbf{u}_{ref}(l)\|$ have been plotted for both load cases and two different meshes (1 and 64 NSRISR elements) in Tables 2 and 3. While for the load case M an analytical solution $\mathbf{u}_{ref}(l)$ is available, we take a reference solution with 8192 first-order Simo–Reissner beam elements according to [6,7] for the load case M + F. Throughout the entire examples section, this specific implementation has been applied wherever a numerical reference solution according to the Simo–Reissner theory is provided. This high number of first-order Simo–Reissner beam elements is necessary in order to gain a higher accuracy than the 64 third-order Kirchhoff elements. In both load cases, we observe a relative error in the range of 2%–3% for 1 NSRISR element and in the range of 10^{-9} for 64 NSRISR elements. Due to the shear forces acting in the load case M + F, we expect a remaining difference between the Kirchhoff and the Simo–Reissner formulation. For this example with a high slenderness ratio of $\rho = 10\,000$, this difference also turned out to be in the range of 10^{-9} . In Examples 2 and 4, this issue will be investigated again in more detail.

In Fig. 5(a), the relative L^2 -error of the load case M is plotted over the element length for different slenderness ratios, when applying the NSRISR element in combination with a full Gauss integration for meshes with 1, 2, 4, 8, 16, 32 and 64 elements. As already indicated by the deformed shapes illustrated in Fig. 2(b) of Section 5, the convergence is slowed down dramatically with increasing slenderness ratio. If the beam is e.g. discretized by one finite element ($l_{ele} = 1000$), the relative error increases almost by two orders of magnitude when enhancing the slenderness ratio from $\rho = 10$ to $\rho = 10\,000$. However, Fig. 5(a) also reveals that this effect decreases with smaller element sizes and almost completely disappears for discretizations with more than 32 elements. The reason for this behavior is the fact that the element slenderness ratio $\rho_{ele} = l_{ele}/r$ is the key-parameter for the observed locking effect and that the latter also decreases with decreasing element sizes. However, for typical engineering applications with relative error bounds of e.g. 1%, the effect is by no means negligible. For sufficiently fine discretizations the expected convergence order of four is reached.

In Fig. 5(b), the relative L^2 -error is plotted for the same slenderness ratios as above when applying the NSRISR element in combination with the MCS method. The locking effect disappears completely for all slenderness ratios. However, as shown in Fig. 5(c) for the slenderness ratio $\rho = 10\,000$, the same effect can be reached for the load case M by a simple reduced integration procedure or a classical ANS approach. On the contrary, Fig. 5(d) reveals that this does not hold for the load case M+F: Both, the ANS approach and the reduced integration scheme can only slightly alleviate the locking effect in the case of rather coarse discretizations. The MCS approach, however, completely eliminates the

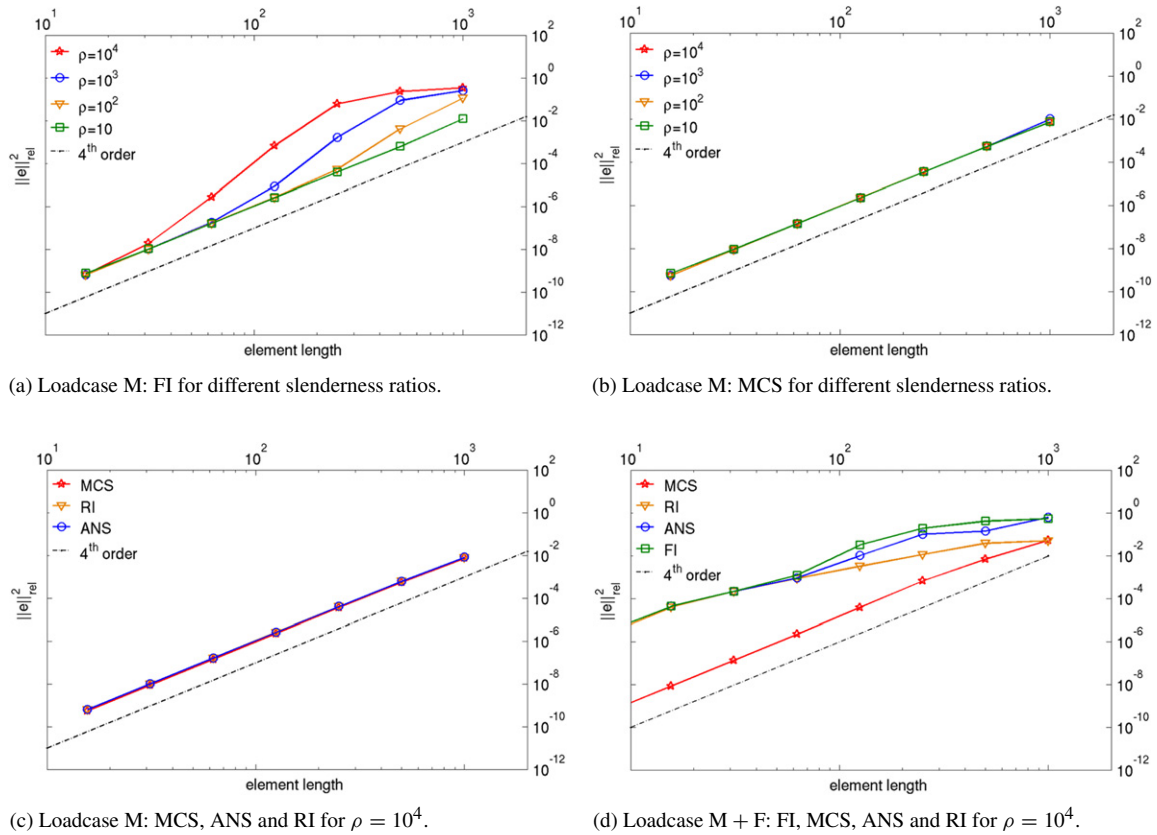


Fig. 5. L_2 -error for the loadcases M and M + F applying the NSRISR element with different integration schemes/anti-locking tools.

error offset due to membrane locking. The explanation for this observation is obvious and in agreement with the statements of Section 5: Similar to the working principle of the MCS method, the reduced integration scheme can alleviate locking effects by reducing the number of constraint equations. But, as shown in Section 5, the MCS method leads to a lower number of constraint equations as compared to the simple reduced integration scheme, which makes the latter method less effective. On the other hand, the pure ANS method is not based on a reduction of constraint equations, but rather on an evaluation of the critical axial tension at selected collocation points with vanishing parasitic strains. As a consequence of the large deformations occurring in geometrically nonlinear beam elements, the positions of these optimal collocation points are deformation-dependent. Obviously, already the relatively simple load case M + F illustrated here lets these positions differ to an extent that almost completely destroys the working principle and impact of the ANS method.

Summing up, it is to say that based on these two examples the new MCS method seems to be superior to standard methods such as reduced integration or ANS in terms of locking avoidance for geometrically nonlinear beam elements. Example 4, also involving 3D deformation states and an initially curved geometry, will confirm this statement.

6.2. Example 2: Slender beam under line load

Let us consider an initially straight beam with slenderness ratio $\rho = 10000$ (thus $r = 0.1$) with support and loading conditions as illustrated in Fig. 6. While the left end of the beam is clamped, the right end only has two remaining degrees of freedom: the displacement u_x in x -direction and the rotation α with respect to the x -axis. Within two steps, the beam is loaded with a constant axial line load and a “rotating” transversal line load according to $\tilde{\mathbf{f}}(s) = (\tilde{f}_x, \tilde{f}_\perp \sin \phi, \tilde{f}_\perp \cos \phi)^T$, with $\tilde{f}_x/EI = \tilde{f}_\perp/(2EI) = 3.6 \cdot 10^{-8}$ and $\phi = 2\pi s/l$. The motivation for the investigation of this example is twofold: On the one hand, this is an example that fulfills the restrictions of all reduced models presented in Sections 3.2 and 3.3 (“isotropic bending”, no external moments, small axial tension due to the

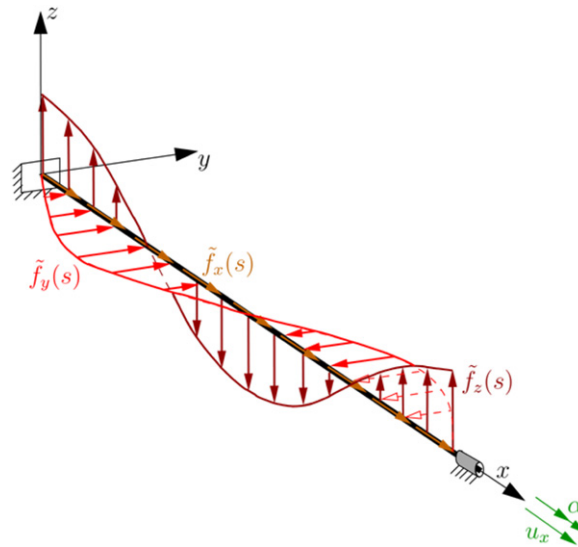


Fig. 6. Initially straight beam supported at both ends and loaded with a “rotating” line load: Initial geometry and external loads.

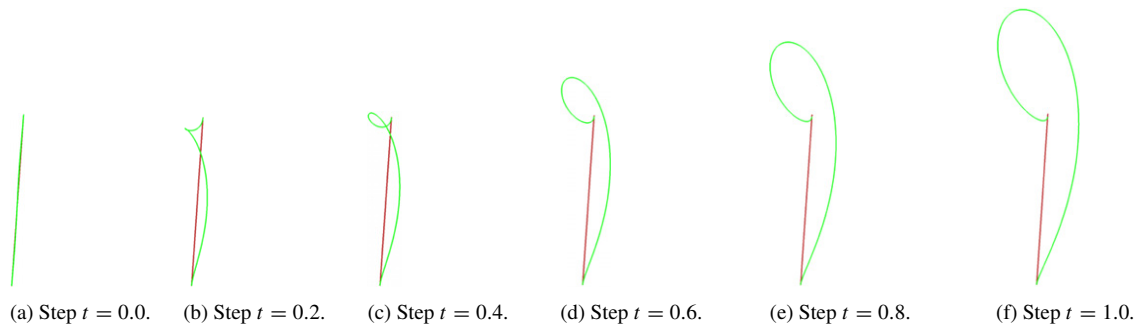


Fig. 7. Beam with a “rotating” line load: Initial and deformed configurations at different load steps.

high slenderness ratio), and which nevertheless leads to a very complex 3-dimensional deformed geometry. Therefore, it is an ideal test case for these formulations. On the other hand, the conditions (concerning the external loads, the very high slenderness ratio and the degree of deformation) prevailing in possible fields of application, e.g. in biopolymer networks (see [40]), are mimicked quite well by this example. The deformed configurations at different load steps are illustrated in Fig. 7.

In Fig. 8(a), the relative L^2 -error of the complete NSRISR element and the reduced formulations is plotted for meshes with 1, 2, 4, 8, 16, 32, 64, 128, 256 and 512 elements with respect to the solution of a sufficiently fine NSRISR discretization. While the complete NSRISR element, the “isotropic” as well as the “isotropic + torsion-free” formulation show no remarkable differences, the formulation based on the “small tension” assumption can apparently not go below a relative error level of $1.0 \cdot 10^{-8}$ for this example. Actually, this is exactly the result we would have expected: While the reduced models according to the “isotropic” and the “isotropic + torsion-free” case are exact formulations in case the corresponding restrictions concerning beam geometry and external loads are fulfilled, the “small tension” assumption can be applied as a sensible approximation also to more general load cases with non-vanishing axial tension. Nevertheless, the exact solution can only be reproduced for load cases leading to zero axial tension (e.g. pure moment loads). However, due to the high slenderness ratio and the resulting small axial strains appearing in application fields as e.g. biopolymer networks, the very simple weak form given in (37) seems to be a very efficient and on the other hand sufficiently accurate simulation tool.

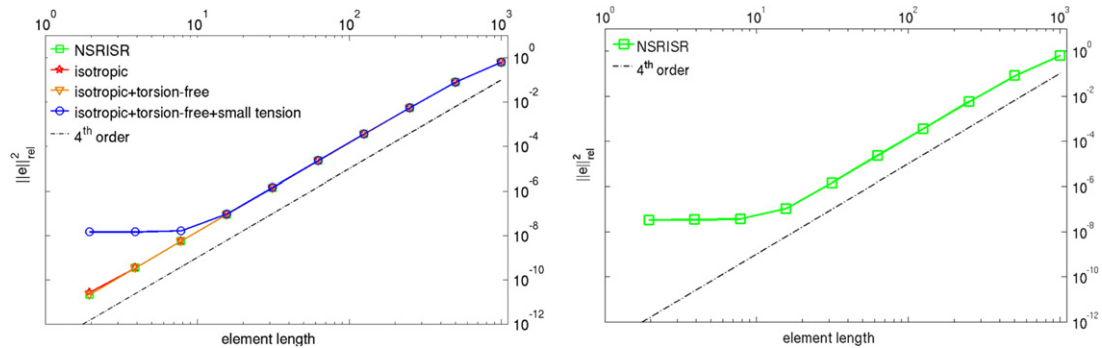
(a) Relative L^2 -error for different element formulations.(b) Relative L^2 -error with “Reissner” reference solution.Fig. 8. Beam with a “rotating” line load: Relative L^2 -errors for different formulations and reference solutions.

Table 4

Beam with a “rotating” line load: midpoint displacement and corresponding relative error for different formulations and meshes.

Formulation	# Elements	$u_x(l/2)$	$u_y(l/2)$	$u_z(l/2)$	$e_{rel}(l/2)$
Reissner	32 768	−5.9992845	−2.3034045	−62.7460017	–
NSRISR	2	−6.3241813	−1.4803128	−61.3713248	$2.6 \cdot 10^{-2}$
NSRISR	128	−5.9992840	−2.3034042	−62.7459997	$3.3 \cdot 10^{-8}$

Additionally, in Fig. 8(b), the relative L^2 -error of the full NSRISR formulation when measured against a Simo–Reissner reference solution consisting of 32 768 first-order beam elements [6,7], is plotted. Again, we observe the correct convergence order of four. The conceptual difference between the shear-deformable Simo–Reissner and the shear-free Kirchhoff beam formulation is visible in form of a remaining constant error level that cannot be undercut even for arbitrarily fine discretizations. This relative error level takes a value of 10^{-7} to 10^{-8} for the presented example and is expected to decrease with increasing slenderness ratio. The displacements and relative errors at the position $x = l/2$ are illustrated in Table 4. The relative errors at this discrete point are in the same range as the corresponding L^2 -errors.

This example also gives some interesting insight into the “one-way” coupling between bending and torsional moments in the geometrically nonlinear regime of isotropic beams: Despite the non-constant bending moment vector $El\kappa(s)$ resulting from the three-dimensional deformed geometry of this example, the torsional moment $m_{\parallel}(s)$ vanishes at any position s along the entire beam. Thus, the bending moment field does not influence the devolution of the torsional moment. On the other hand, one can show that an additional external moment $\bar{\mathbf{m}} = (\bar{m}_{\parallel}, 0, 0)^T$ in global x -direction applied at the beams right end would lead to a constant torsional moment $m_{\parallel}(s) = \bar{m}_{\parallel}$ along the entire beam and would change the deformed centerline curve. This means that the torsional moment indeed influences the devolution of the bending moment. The term $\mathbf{h}_3$ of the weak form (32) can be identified as the corresponding coupling term.

6.3. Example 3: Straight beam bent to helix

The next example is an initially straight, clamped beam with a slenderness ratio $\rho = 100$ ($r = 10$) loaded by the end-moment $\mathbf{M} = (M, 0, M)^T$, with $M = 10.0$. Due to the complex deformation path involving large 3D translations and rotations, the load has been applied in a total of 400 steps. The example represents an isotropic beam with non-vanishing torsion and serves as test case for the “isotropic bending” formulation of Section 3.2.1. Furthermore, an analytical solution can be derived for this load case by means of the differential equations given in Section 2.4. The initial and deformed configuration of this problem setting are illustrated in Fig. 9(a). One can verify that the differential equations (24) and (25) presented in Section 2.4 as well as the Dirichlet- and Neumann boundary conditions of this

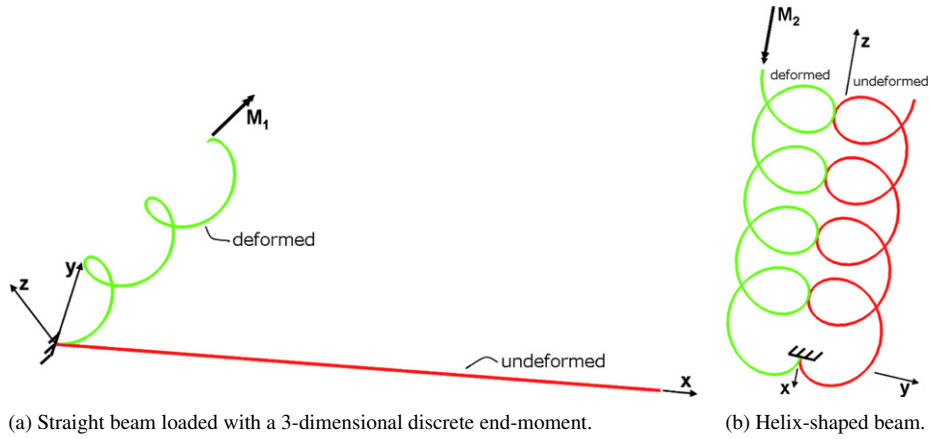


Fig. 9. Straight beam bent to a helix and inverse problem of a helix-shaped beam bent to an inverse helix.

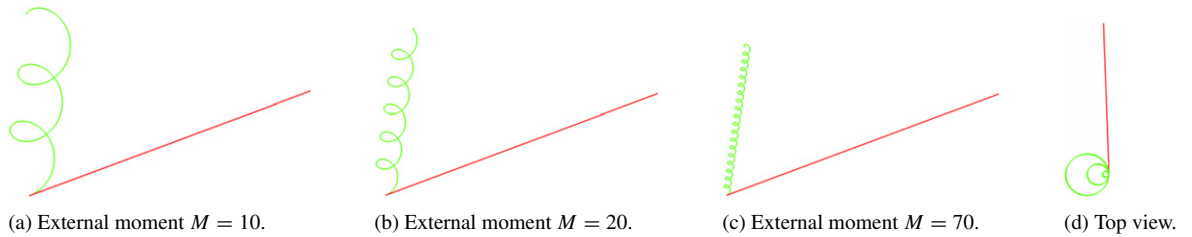


Fig. 10. Deformed shapes of an initially straight beam for different magnitudes of the applied 3-dimensional moment.

example are fulfilled by an analytical solution which is characterized by the following representation:

$$\mathbf{r}(s) = R_0 \begin{pmatrix} \frac{1}{\sqrt{2}} (\sin \beta + \beta) \\ \frac{1}{\sqrt{2}} (1 - \cos \beta) \\ \frac{1}{\sqrt{2}} (\beta - \sin \beta) \end{pmatrix} \quad \text{with } R_0 = \frac{EI}{2M} \quad \text{and} \quad \beta = \frac{s}{\sqrt{2}R_0}. \quad (59)$$

This solution represents a helix whose mid-axis points into the $(1, 0, 1)^T$ -direction, which is the direction of the applied external moment. The special parameter choice of this example leads to a radius R_0 of the enveloping cylinder that is identical to the slope of the helix. In Fig. 10, the deformed shapes of the beam are illustrated for different magnitudes of the external moment.

In the following, we want to compare this example with the inverse problem as shown in Fig. 9(b). For the inverse problem, the initial geometry is already a helix according to (59). However, for this example, a different radius $\tilde{R}_0 = l/(9\sqrt{2}\pi) < R_0$, leading to 4.5 loops along the standard length $l = 1000$, has been chosen. Since the length of the beam still has the standard value of $l = 1000$, a higher number of loops along the helix results from this choice. Again, one can verify by means of the differential Eqs. (24) and (25) that the external moment $\tilde{\mathbf{M}} = (-\tilde{M}, 0, -\tilde{M})^T$ with $\tilde{M} = EI/(2\tilde{R}_0)$ exactly bends the helix into a straight line (see Fig. 11(c)). By doubling the external moment to $\tilde{M} = EI/\tilde{R}_0$ (applied in 1000 load steps), the helix is exactly inverted, i.e. the analytical solution for the deformed geometry has the same form as for the initial geometry according to (59) (with $\tilde{R}_0$ instead of R_0), but with opposite sign in the second vector component. Different deformation states for the latter case are illustrated in Fig. 11.

Although the two cases “straight line bent to helix” and “helix bent to straight line” seem to be very similar at first glance, there is a crucial difference from a mechanical point of view: In the former case, the initial curvature vanishes, which enables the application of the reduced formulation according to the “isotropic bending” case. In the latter case, the straight configuration only represents one special deformed state. In general, the initial curvature has to be considered, which requires the application of the complete NSRISR formulation as derived in Section 2. In

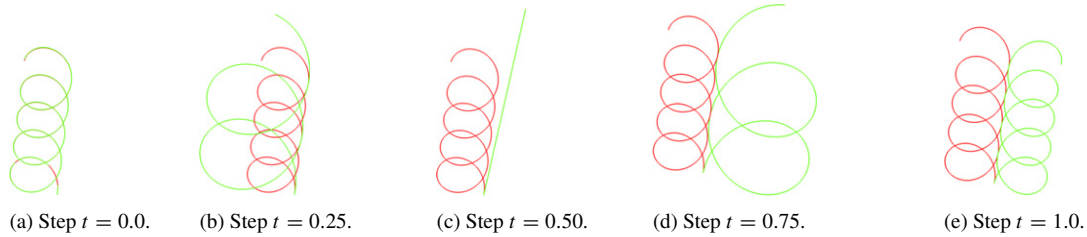


Fig. 11. Inversion of a helix by an external end-moment: Deformed shape at different load steps.

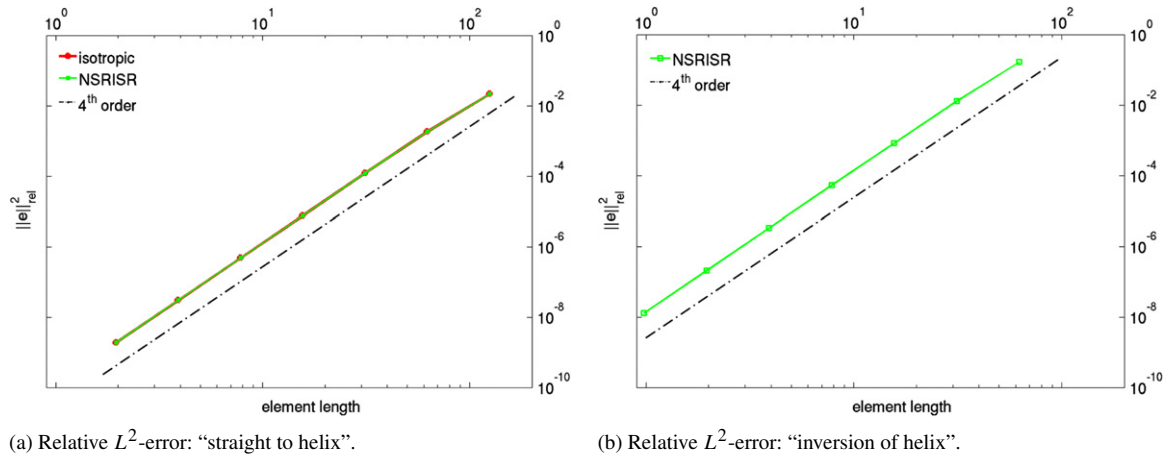


Fig. 12. Convergence plot of load cases “straight to helix” and “inversion of helix”.

Table 5

Loadcase ‘straight to helix’: tip displacement and corresponding relative error for different formulations and meshes.

Formulation	# Elements	$u_x(l)$	$u_y(l)$	$u_z(l)$	$e_{rel}(l)$
Analytical	–	–528.0745578	54.3058724	528.0745578	–
NSRISR	8	–511.5332077	77.8086301	523.8770652	$3.9 \cdot 10^{-2}$
NSRISR	512	–528.0745568	54.3058748	528.0745579	$3.5 \cdot 10^{-9}$

Fig. 12(a), the relative L^2 -error of the case “straight line bent to helix” is plotted for the complete NSRISR as well as for the reduced “isotropic” formulation and meshes with 8, 16, 32, 64, 128, 256 and 512 elements. Again, the reduced formulation exactly represents the solution of the complete formulation.

Fig. 12(b) illustrates the corresponding L^2 -error of the case “inversion of helix” for the complete NSRISR element and meshes with 16, 32, 64, 128, 256, 512 and 1024 elements. For both load cases, analytical solutions according to (59) have been applied as reference. For all formulations and load cases, the investigated element formulation converges towards the analytical solution with the expected optimal order of four. In an arguably more illustrative way, the results of these two load cases are presented in terms of pointwise displacements and relative errors at the tip of the beam in Tables 5 and 6. As already mentioned above, the initial and deformed configurations of the load case “inversion of helix” are symmetric to the xz -plane. Consequently, the analytical solution for the tip displacement exhibits only a component in y -direction. Again, the pointwise error levels are in agreement with the L^2 -errors.

This example also clarifies the relevance of the Frenet–Serret reference triad for analytical purposes, e.g. for the derivation of analytical solutions via (24) and (25). On the one hand, helical curves represent a distinguished class of solutions for beams loaded with moments (see e.g. [1]). On the other hand, this kind of solutions can easily be represented with respect to the Frenet–Serret reference triad, since the curvature $\kappa_{helix} = const.$ as well as the torsion $\tau_{FS, helix} = const.$ of the reference triad are constant in this case.

Table 6

Loadcase ‘inversion of helix’: tip displacement and corresponding relative error for different formulations and meshes.

Formulation	# Elements	$u_x(l)$	$u_y(l)$	$u_z(l)$	$e_{rel}(l)$
Analytical	–	0.0	–100.0351462	0.0	–
NSRISR	16	10.4605821	–85.6543599	–21.0312949	$2.8 \cdot 10^{-1}$
NSRISR	1024	0.0000012	–100.0351464	–0.0000019	$2.3 \cdot 10^{-8}$

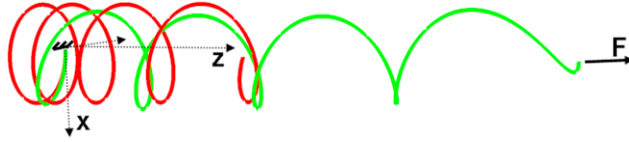


Fig. 13. Axial force acting on a helix: Problem setup and deformed configuration.

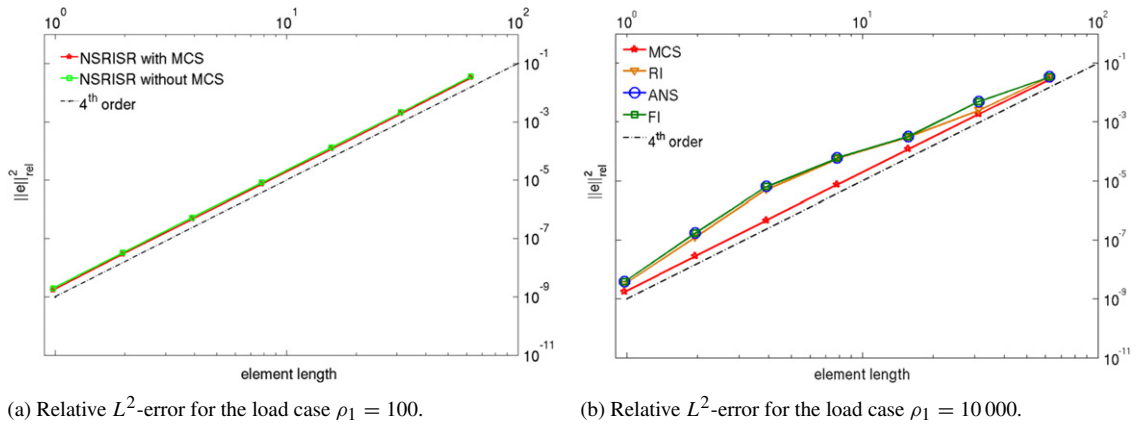


Fig. 14. Axial force acting on a helix: Convergence behavior for different slenderness ratios with and without MCS method.

6.4. Example 4: Axial force acting on a helix

Finally, we want to present a more complex example with 3D initial geometry as illustrated in Fig. 13 (see also [39]). A helix with linearly increasing slope, clamped at its right end, is loaded with a discrete end-force $\mathbf{F} = (0, 0, F)^T$ at its left end. The initial geometry of the helix can be described by the following parametrized space curve:

$$\mathbf{r}_0(\beta) = R_0 \begin{pmatrix} \sin \beta \\ \cos \beta - 1 \\ \frac{6}{81\pi^2} \beta^2 \end{pmatrix}$$

with $R_0 = \frac{l}{6\sqrt{\left(\frac{3\pi}{4}\right)^2 + 1 + \frac{27\pi^2}{8} \ln\left(\frac{4}{3\pi} + \sqrt{1 + \left(\frac{4}{3\pi}\right)^2}\right)}} \approx 34.36$ and $\beta \in [0; 9\pi]$. (60)

The radius R_0 is chosen such that the helix exactly consists of 4.5 loops along the standard length $l = 1000$. This example has been simulated for the two different slenderness ratios $\rho_1 = 100$ (6 load steps) and $\rho_2 = 10000$ (12 load steps) in order to examine the locking behavior of the element in 3D scenarios. Accordingly, we have chosen the values $F_1 = 2.0 \cdot 10^{-1}$ and $F_2 = 2.0 \cdot 10^{-9}$ for the two different slenderness ratios, respectively. The ratio of these forces is chosen identical to the ratio of the bending stiffnesses of the cases $\rho_1 = 100$ and $\rho_2 = 10000$ leading to the comparable values $u_{z,max,1} \approx 267$ and $u_{z,max,2} \approx 266$ for the maximal tip-displacement in z -direction.

In Fig. 14(a), the resulting relative L^2 -error for the NSRISR element with and without MCS approach is plotted for the case $\rho = 100$ and meshes with 16, 32, 64, 128, 256, 512 and 1024 elements. In the case of the higher slenderness

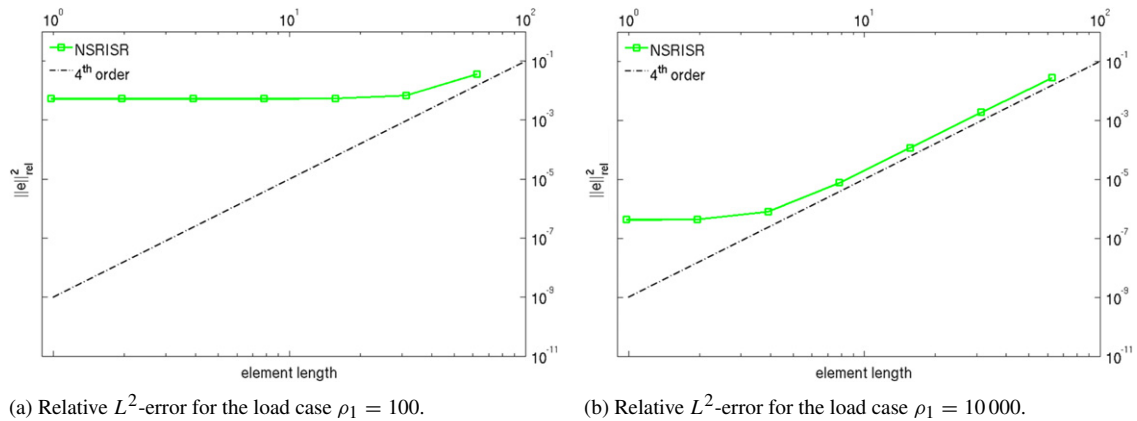


Fig. 15. Axial force acting on a helix: Convergence behavior for different slenderness ratios with “Simo–Reissner” reference solution.

ratio $\rho = 10000$, which is more critical with respect to membrane locking, we again compare the different methods MCS (corresponding to the case “NSRISR with MCS” in Fig. 14(a)), ANS, RI and FI (corresponding to the case “NSRISR without MCS” in Fig. 14(a)) as already done in Example 1. The corresponding L^2 -errors are plotted in Fig. 14(b).

The MCS variant reaches almost identical L^2 -error levels and an optimal convergence order of four for both, the moderate slenderness ratio of $\rho_1 = 100$ and the extreme slenderness ratio of $\rho_2 = 10000$. While the original element formulation without MCS approach and full Gauss integration produces comparable results for the first load case, it exhibits an error offset due to membrane locking for the highly slender geometry. The methods ANS and RI hardly improve this behavior, thus again demonstrating the superiority of the MCS approach for such extremely high slenderness ratios.

However, the error offset between these two slenderness ratios is lower than in the 2D-example of Section 6.1. This observation confirms the prediction of Section 5 that membrane locking may be less pronounced for 3D problems. In conclusion, it can be stated that the NSRISR element formulation of Section 2 in combination with the MCS approach of Section 5 completely remedies the unphysical effects of membrane locking even for extreme slenderness ratios and demanding geometries.

In Fig. 15(a) and (b) the relative L^2 -errors of the load cases and meshes investigated above are plotted for the NSRISR element including the MCS approach with respect to a Simo–Reissner beam reference solution [6,7] based on meshes with 1032 (for $\rho = 100$) and 32 768 (for $\rho = 10000$) elements. Due to the shear forces prevailing in this example, we again observe a kink in the error evolution, which represents the model error existing between Kirchhoff and Simo–Reissner formulations. As expected, this model error decreases with increasing slenderness ratio from approximately 1% for $\rho = 100$ to approximately 10^{-6} for $\rho = 10000$. The corresponding tip displacements and errors presented in Tables 7 and 8 confirm these results.

7. Conclusion

In our earlier work [39], a geometrically exact, objective, large-deformation finite element formulation for thin Kirchhoff rods accounting for arbitrarily curved initial geometries has been proposed. The core component of this formulation is the definition of a reference triad field, whose proper interpolation is substantial in order to derive objective strain measures being invariant under rigid body motions as well as path-independent. Crucial topics, such as objectivity and path-independence of the discrete strain measures, the choice of suitable test and trial functions as well as the investigation of possible reference triad definitions have already been addressed in [39].

In the current contribution, we have particularly emphasized the important issue of membrane locking. While in [39], we predominantly investigated beams of moderate slenderness, a regime in which membrane locking plays only a minor role, we have focused on beams with extreme slenderness ratios within the current work. Since the effect of membrane locking is not negligible anymore in this case, we have proposed a new method, denoted as Minimal Constraint Strains (MCS), in order to effectively avoid this phenomenon. The proposed method comes with

Table 7

Case $\rho = 100$: tip displacement and corresponding relative error for different formulations and meshes.

Formulation	# Elements	$u_x(l)$	$u_y(l)$	$u_z(l)$	$e_{rel}(l)$
Reissner	1024	6.9880795	34.9696523	269.2174165	–
NSRISR	16	6.9611702	34.8046848	254.8547030	$5.3 \cdot 10^{-2}$
NSRISR	32	6.9662594	34.9371216	266.2444443	$1.1 \cdot 10^{-2}$

Table 8

Case $\rho = 10\,000$: tip displacement and corresponding relative error for different formulations and meshes.

Formulation	# Elements	$u_x(l)$	$u_y(l)$	$u_z(l)$	$e_{rel}(l)$
Reissner	32 768	6.9689997	34.9556163	266.2335940	–
NSRISR	16	6.8968737	34.8144509	255.3321931	$4.1 \cdot 10^{-2}$
NSRISR	512	6.9690134	34.9556162	266.2333813	$7.9 \cdot 10^{-7}$

low numerical costs and it is expected that its very general nature enables a simple transferability to other locking phenomena. We have applied the new method to some carefully selected numerical examples and could show its advantages when compared with classical approaches such as Assumed Natural Strains (ANS) or reduced integration.

Besides a comprehensive model allowing for arbitrarily loaded beams of any initial curvature and cross-section shape, we have also derived several reduced beam models, i.e. formulations which are only valid for special problem classes concerning the beam geometry and the external loads. The reduced models are characterized by simplified element formulations, less degrees of freedom and a higher computational efficiency. These properties make them particularly attractive for fields of application with a large number of degrees of freedom and/or long physically relevant time scales, as typical for systems such as biopolymer networks or high-tensile fabric layers in composite materials.

With several numerical examples, we have confirmed the accuracy of the comprehensive and the reduced beam models and the effectivity of the MCS method when applied to highly slender beam structures. Due to the C^1 -continuous centerline representation, the presented beam element also seems to provide an ideal basis for a robust beam-to-beam contact formulation. The application of the proposed formulation to large systems of beams with extreme slenderness ratios and complex contact interaction will be addressed in future work.

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